

ARTICLE

A transcontinental experiment elucidates (mal)adaptation of a cosmopolitan plant to climate in space and time

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Abstract

Climate change and the global spread of non-native species are two of the most significant threats to biodiversity and ecosystem function. Both these phenomena subject populations to novel conditions, either in space (species introductions) or in time (climate change), yet the role of adaptation in how populations respond to these rapid environmental shifts is poorly understood. We conducted a large-scale transcontinental common garden experiment using white clover (*Trifolium repens*, Fabaceae) to test whether adaptive evolution to spatiotemporal variation in climate could contribute to the ecological success of one of the most widespread plant species in the world. Individuals from 96 populations of *Trifolium repens* (white clover) from both its native (Europe) and introduced (North America) ranges were planted into four experimental common gardens located in northern (Uppsala, Sweden) and southern (Montpellier, France) Europe, and northern (Mississauga, Canada) and southern (Lafayette, USA) North America. We recorded plant sexual and clonal fitness in each common garden, and assessed whether the strength of local adaptation differed between the native and introduced ranges, and whether populations are rapidly adapting to climate change. Results show that local adaptation was only evident when populations were transplanted into common gardens located in the same range (native or introduced) from which they

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originated, and was driven by stronger selection (due to climatic factors rather than herbivory) at lower latitudes in both ranges. Our results indicate rapid local adaptation across a large latitudinal gradient in introduced *T. repens* populations, along with an associated adaptation cost when transplanted back into the native range. We also find evidence of an adaptation lag in the northern common garden in the introduced range, with plants from historically warmer climates exhibiting the greatest fitness. These findings support two major conclusions: (1) white clover can rapidly adapt to spatial variation in climate in its introduced range as well as its native range, and (2) despite rapid adaptation to novel environments, introduced white clover populations are not keeping pace with rapid climate change. Overall, our results provide insight into the role of adaptation in facilitating the ecological success of non-native species in a rapidly changing world.

KEYWORDS

adaptation lag, climatic distance, common garden, cyanogenesis, environmental variation, introduced range, local adaptation, native range, non-native species, provenance study, *Trifolium repens*, white clover

INTRODUCTION

Natural selection results in populations evolving traits that are advantageous to the environmental conditions of a particular location, a phenomenon known as local adaptation (Hereford, 2009; Kawecki & Ebert, 2004; Turesson, 1922). A fundamental aspect of local adaptation is that variation in biotic and abiotic factors leads to changes in the relative fitness of genotypes between environments (Gillespie & Turelli, 1989; Kawecki & Ebert, 2004; Lee & Mitchell-Olds, 2011). Such genotype-by-environment interactions can be elucidated by reciprocal transplant experiments, which are challenging to conduct, but vital in demonstrating fitness trade-offs, whereby genotypes exhibit higher relative fitness in their local environment and lower fitness in other environments (Gillespie & Turelli, 1989; Hereford, 2009; Leimu & Fischer, 2008; Savolainen et al., 2013). Human activities are exposing populations to rapidly changing novel environments, both across space through species introductions (Pyšek et al., 2020), and over time through anthropogenically mediated climate change (Hoffmann & Sgrò, 2011). Understanding how populations simultaneously adapt to novel biotic and abiotic environments over space and time is an important problem that is poorly resolved (Hendry, 2017; Siepielski et al., 2017). Addressing this problem is necessary to determine how increasingly pervasive anthropogenic disturbances alter the ecology and evolution of native and introduced species (Fugère & Hendry, 2018; Parmesan, 2006).

Introduced populations are frequently subjected to shifts in selection pressure compared to the native environment

to which they are locally adapted (Colautti & Lau, 2015; Erfmeier, 2013; Lau & terHorst, 2015). This shift in selection pressure in either the biotic environment (e.g., novel species interactions) or the abiotic environment (e.g., climatic factors) could inhibit the establishment and spread of an introduced population (Colautti & Lau, 2015; Erfmeier, 2013; Lau & terHorst, 2015). Additionally, inherent characteristics of introduced populations, such as small population size and low genetic diversity, could limit their ability to respond to novel selection pressures, with Allee effects and genetic bottlenecks potentially resulting in mate limitation and inbreeding depression (Allendorf & Lundquist, 2003; Burns et al., 2011; Estoup et al., 2016; Kolar & Lodge, 2001; Lau & terHorst, 2015; Pannell et al., 2015; Taylor & Hastings, 2005; Tsutsui et al., 2000). In most cases, the ecological and/or genetic constraints experienced by introduced populations are sufficient to result in the failure of those populations to establish and their eventual extirpation (Jeschke & Pyšek, 2018; Kolar & Lodge, 2001).

Despite the challenges presented by introduction to novel environments across space, introduced populations still often become invasive, experiencing substantial ecological opportunity if constraints can be overcome. For example, ecological constraints can be overcome if novel selection regimes in introduced environments are either benign or even favorable to a non-native population (Colautti & Lau, 2015; Erfmeier, 2013). This is particularly true when an introduced population has an advantage over its new natural enemies or competitors compared to in its native range (Blossey & Nötzold, 1995; Callaway et al., 2022; Callaway & Ridenour, 2004;

Müller-Schärer et al., 2004). It is also possible that introduced populations do not exhibit strong evidence of genetic constraints to begin with (i.e., the genetic paradox of invasions), because any initial reductions in genetic variation are not substantial enough to limit the ability of a population to respond to novel selection pressures (Allendorf & Lundquist, 2003; Battlay et al., 2025; Colautti & Barrett, 2013; Dlugosch & Parker, 2008a, 2008b; Estoup et al., 2016; Maron et al., 2004; van Boheemen et al., 2017). In fact, when introduced populations experience strong selection and sufficient genetic variation in novel environments, it could facilitate rapid adaptation to those novel environments, which could in turn, contribute to the success of a non-native species (Colautti & Barrett, 2013; Colautti & Lau, 2015; Dlugosch & Parker, 2008a; Maron et al., 2004; van Boheemen et al., 2019; Simón-Porcar et al., 2021; but see Genton et al., 2005). However, the extent to which rapid local adaptation to novel biotic and abiotic conditions can facilitate the success of a widespread non-native species across large environmental gradients remains poorly understood.

While there are examples in which introduced populations have rapidly adapted to novel environments across space, it is important to consider that these novel environments may also be changing through time. Temporal shifts in a local environment can cause selection pressures to fluctuate, which could result in traits that were previously adapted to that environment becoming maladaptive (Aguirre-Liguori et al., 2019; Anderson & Wadgymar, 2020; Bontrager et al., 2020; Capblancq et al., 2020; Exposito-Alonso et al., 2019; Wilczek et al., 2014). If the rate of adaptation is exceeded by the rate of environmental change, this disruption of local adaptation is likely to result in reduced fitness (i.e., maladaptation) of populations (Aguirre-Liguori et al., 2019; Atkins & Travis, 2010; Capblancq et al., 2020; St Clair & Howe, 2007; Wilczek et al., 2014). Recent global warming may result in populations becoming maladapted to their local climate conditions, whereas populations from lower latitudes or elevations (i.e., warmer origins) become better suited to environmental conditions at higher latitudes or elevations; a phenomenon known as an adaptation lag (Anderson & Wadgymar, 2020; Kooyers et al., 2019; Wilczek et al., 2014). Currently, the frequency and magnitude of adaptation lags are poorly understood (Anderson & Song, 2020).

If non-native species rapidly adapt to novel environments across space, similar rapid adaptation to climate change through time could facilitate their continued range expansion, persistence, and pervasiveness (Byers, 2002; Colautti & Barrett, 2013; Maron et al., 2004; Oduor et al., 2016; Walther et al., 2009). In fact, non-native

species are often better at climate tracking and more resistant to climate warming than native species, which could contribute to their success as invaders (Wolkovich et al., 2013). However, the interplay between local adaptation, species introductions, and anthropogenically mediated climate change remains an understudied, yet critically important, aspect of understanding the ongoing human-induced biodiversity crisis (Brondizio et al., 2019). This problem is best addressed through large field experiments testing local adaptation in both the native and introduced ranges of widespread non-native species, alongside elucidating the environmental conditions that may result in rapid adaptation and success in a novel environment (Anderson & Song, 2020; Colautti & Lau, 2015).

Plant-herbivore interactions have been highlighted as key selective factors that determine patterns of local adaptation, species responses to environmental change, and biological invasions. Spatial variation in both biotic and abiotic factors is likely to lead to spatial variation in plant defense traits by affecting the costs and benefits involved in growth-defense trade-offs (Anstett et al., 2018; Baskett & Schemske, 2018; Moreira et al., 2018). Local herbivore communities impose selection through preferential feeding on plants with lower resistance, influencing the benefits of plant defenses in areas of particularly high herbivore pressure (Agrawal, 2007; Coley & Barone, 1996; Ehrlich & Raven, 1964; Johnson & Rasmann, 2011; Züst et al., 2012). In turn, the metabolic costs of defenses can be substantial, while abiotic environments and the stressors they impose can influence the availability of resources to plants, exacerbating defense costs (Coley et al., 1985; Fine et al., 2006; Hahn et al., 2019; Herms & Mattson, 1992). Furthermore, it is increasingly recognized that anthropogenic activity can influence the evolution of plant defenses (Bidart-Bouzat & Imeh-Nathaniel, 2008; Massad & Dyer, 2010; Miles et al., 2019). Climate warming could result in increased herbivore feeding and/or increased plant defenses (Dostálek et al., 2020; Lemoine et al., 2013) or could contribute to the success of a non-native species by decoupling locally adapted plant-herbivore interactions among native species (Adams et al., 2009; Costan et al., 2022; but see Agrawal & Kotanen, 2003). It is therefore important to understand how populations are adapting to their local biotic and abiotic environments and how anthropogenic factors can affect adaptation.

An ideal study system to test questions surrounding adaptation to novel environments post-introduction and in response to climate change would be a widespread species with known agents of biotic and abiotic selection across space and over time that can drive adaptation in its native and introduced ranges. *Trifolium repens* is an allopolyploid perennial legume that is native to Eurasia

and now has a cosmopolitan distribution, following its introduction to inhabited continents throughout the world, including North America (Daday, 1958; Griffiths et al., 2019; Kjærgaard, 2003). Mechanisms of biotic and abiotic selection in *T. repens* have been studied for over 70 years. In *T. repens*, clines in a prominent chemical defense mechanism, hydrogen cyanide (HCN), have been identified across many types of environmental gradients (Daday, 1954a, 1954b, 1958; Kooyers et al., 2014; Kooyers & Olsen, 2013; Thompson et al., 2016; Till, 1987), providing opportunities to test hypotheses surrounding the involvement of biotic and abiotic factors in adaptation to environmental change.

To understand adaptation to environmental variation in a non-native species over space and time, we conducted a transcontinental common garden field experiment across latitudinal gradients in both the native (Europe) and introduced (North America) ranges of *T. repens*. We first address four research questions focused on adaptation in *T. repens* to biotic and abiotic environmental variation in space. (Q1) Are *T. repens* populations locally adapted to their climate of origin and does the strength of local adaptation differ across space? This question is a prerequisite before any comparisons can be made regarding local adaptation between the native and introduced ranges. (Q2) Does *T. repens* exhibit stronger local adaptation in its native range than in its introduced range? (Q3) What are the roles of plant defense and herbivory (i.e., biotic factors) in driving local adaptation across space in *T. repens*? (Q4) What bioclimatic variables (i.e., abiotic factors) have the strongest impact on fitness across space (latitudes and ranges) in *T. repens*? We then focus on whether *T. repens* can adapt to climate change through time using a space-for-time approach to analyze a subset of the common garden experimental data (Fournier-Level et al., 2011; Pickett, 1989), asking: (Q5) Are there adaptation lags to climate change in *T. repens*, such that populations originating from warmer climates exhibit greater fitness than local populations? This study is among the first to investigate the ability of an introduced species to rapidly adapt to novel environments in space and time simultaneously. Our results elucidate the potential for rapid adaptation to contribute to initial and continued ecological success of non-native species in the face of ongoing environmental change.

METHODS

Study system

Trifolium repens has been a model for the study of plant defense based on its polymorphism for the production of

the potent chemical defense HCN (Hughes, 1991). In fact, the ability to produce HCN as an antiherbivore defense may have contributed to the range expansion of *T. repens*, allowing it to thrive in human-disturbed habitats such as lawns, fields, and roadsides in many regions throughout the world (Burdon, 1983; Kjærgaard, 2003; Olsen et al., 2021). Cyanogenesis occurs through the hydrolyzation of cyanogenic glucosides by the enzyme linamarase. In *T. repens*, the presence or absence of cyanogenic glucosides, linamarin and lotaustralin, is controlled by three linked genes in their metabolic pathway (*CYP79D15*, *CYP736A187*, and *UGT85K17*; referred to collectively as *Ac/ac*), while the presence or absence of the enzyme linamarase is controlled by the gene *Li/li* (Olsen et al., 2007; Olsen & Small, 2018). Thus, four phenotypes exist for HCN and the two metabolic components required for its production: one in which HCN is constitutively produced (*AcLi*) and three that are acyanogenic and, therefore, lack the defense entirely (*acLi*, *Acli*, *accli*; Atwood & Sullivan, 1943; Corkill, 1940).

Cyanogenesis in *T. repens* is widely considered an adaptive polymorphism because it exhibits clines in the presence of HCN across numerous types of environmental gradients (Brighton & Horne, 1977; Daday, 1954a, 1954b, 1958; Innes et al., 2022; Santangelo et al., 2022; Thompson et al., 2016). In general, dominant *Ac* and *Li* alleles (and thus the ability to produce HCN) are fixed in warmer climates, while recessive *ac* and *li* alleles are fixed in colder climates (Daday, 1954a, 1954b, 1965). Latitudinal cyanogenesis clines have been identified in the native European range of *T. repens* (Daday, 1954a), as well as its introduced ranges, such as North America (Daday, 1965). However, clines within introduced ranges are generally shallower, possibly indicating weaker local adaptation post-introduction, which occurred less than 400 years ago in the case of North America (Daday, 1958; Innes et al., 2022; Kjærgaard, 2003; Kooyers & Olsen, 2012). One hypothesis to explain cyanogenesis clines is a gradient in herbivory, with greater diversity and abundance of herbivores in warmer climates resulting in stronger selection favoring cyanogenic plants (Angseesing, 1974; Dirzo & Harper, 1982; Kakes, 1989). Alternatively, empirical studies across various environmental gradients have identified temperature, soil moisture, and soil nutrients as abiotic factors that may impose selection on cyanogenesis or its metabolic components to influence cline formation (Daday, 1954a, 1965; Gleadow & Woodrow, 2002; Kooyers et al., 2014, 2018). Overall, despite 70 years of research on the drivers of cyanogenesis clines in *T. repens*, there remains little consensus on how variation in environmental conditions influences selection on cyanogenesis in *T. repens*, or if cyanogenesis is primarily responsible for rapid local

adaptation to novel environments (Albano & Johnson, 2023; Daday, 1954a, 1965; Hughes, 1991; Kooyers et al., 2014; Kooyers & Olsen, 2013; Kuo et al., 2024; Wright et al., 2018, 2022). Our study aims to resolve this uncertainty by directly testing the strength of local adaptation and its biotic and abiotic drivers in both the native and introduced ranges of *T. repens*.

Sampled populations and common garden sites

We addressed our research questions using four common gardens distributed across northern (Uppsala, Sweden) and southern (Montpellier, France) regions of the native European range of *T. repens*, and northern (Mississauga, ON, Canada) and southern (Lafayette, LA, USA) regions of the North American introduced range (Figure 1). Seeds were collected from 49 natural *T. repens* populations scattered across the native European range spanning $\sim 27^\circ$ of latitude, and 47 natural *T. repens* populations along a 2500 km north–south transect spanning $\sim 21^\circ$ of latitude in the introduced North American range (Figure 1). Temperature and latitude are highly correlated in eastern North America but less directly associated in Europe, with temperatures varying across both north–south and east–west gradients, which led to the more diffuse sampling approach in Europe and the transect sampling approach in North America. To minimize maternal and epigenetic effects, we hand-pollinated plants germinated from field-collected seeds in a growth chamber environment (25°C day, 15°C night, 14 h:10 h light:dark cycle; Conviron MTPS, Winnipeg, MB, Canada) to create an outbred F1 cohort of seeds (i.e., a refresher generation) for each population. F1 generation seeds were germinated and established for 4 weeks in peat pots (8 cm diameter, 591 mL volume) in uniform growth chamber or greenhouse conditions near each common garden site. Five replicate individuals from each population were transplanted in peat pots directly into the natural soil of a typical cultivated lawn at each common garden site. Inconsistent germination success resulted in some populations not being included in some common garden locations, and in very limited cases, populations had four or six individuals planted. The resulting total number of plants per garden was 465, 465, 458, and 470 in Mississauga, Lafayette, Uppsala, and Montpellier, respectively. Plants were arranged in rows and columns with 1 m spacing, and black landscaping fabric was used to cover the entirety of the lawn to prevent the rooting of stolons into the natural soil between plants. Holes (12 cm diameter) were cut in the fabric for each individual plant, leaving a 2 cm

ring of natural vegetation around each plant to allow for natural competition. This vegetation was trimmed monthly throughout the experiment. Planting dates matched growing seasons in each location. The Lafayette and Mississauga common gardens were established in April and May 2020, respectively, while the Montpellier and Uppsala common gardens were postponed due to COVID-19 restrictions in Europe, leading to their establishment in March and May 2021, respectively. Plants were given supplemental water for the first 2 weeks of the experiment and intermittently after the first 2 weeks when necessary to avoid excessive mortality. Plants that died within the first 2 weeks of the experiment were deemed to have died due to the compounding effects of transplantation stress with short-term stressful climatic conditions, such as high temperature and drought at each garden, and were replaced using extra germinated individuals from the same population (Mississauga: 3; Lafayette: 32; Uppsala: 72; Montpellier: 84). Plants that died after 2 weeks were deemed to have died solely due to the local environmental conditions at each common garden site post-establishment and were not replaced. Each common garden was maintained for two full growing seasons.

Tissue collection, herbivory estimates, and measures of fitness

After 1–2 months of growth, one to three fully expanded leaves were harvested from each surviving individual and frozen at -20°C . These leaves were later freeze-dried and used for DNA extraction, followed by low coverage whole genome sequencing to predict the presence of *Ac/ac* and *Li/li* alleles (see Appendix S1: [Supplementary Methods](#)). In total, 586 individuals (Mississauga: 302, Lafayette: 99, Uppsala: 78, Montpellier: 107) were successfully sequenced and characterized as possessing either dominant or recessive alleles at *Ac/ac* and *Li/li*. The remaining 1272 individuals were not genotyped because they either died before tissue could be harvested or genomic library preparations were unsuccessful. We exposed plants to the natural environment of each common garden at the earliest life stage possible to capture potentially important selection on seedlings (Agrawal et al., 2012). As a result, the experiment experienced high mortality of young individuals, which is common and a natural part of the biology of *T. repens* and other plants (Agrawal et al., 2012; Escudero et al., 2000; Moles & Westoby, 2004). This mortality prevented many individuals from being included in tissue collection.

Visual herbivory estimates (% leaf area consumed) were conducted monthly following the methods of

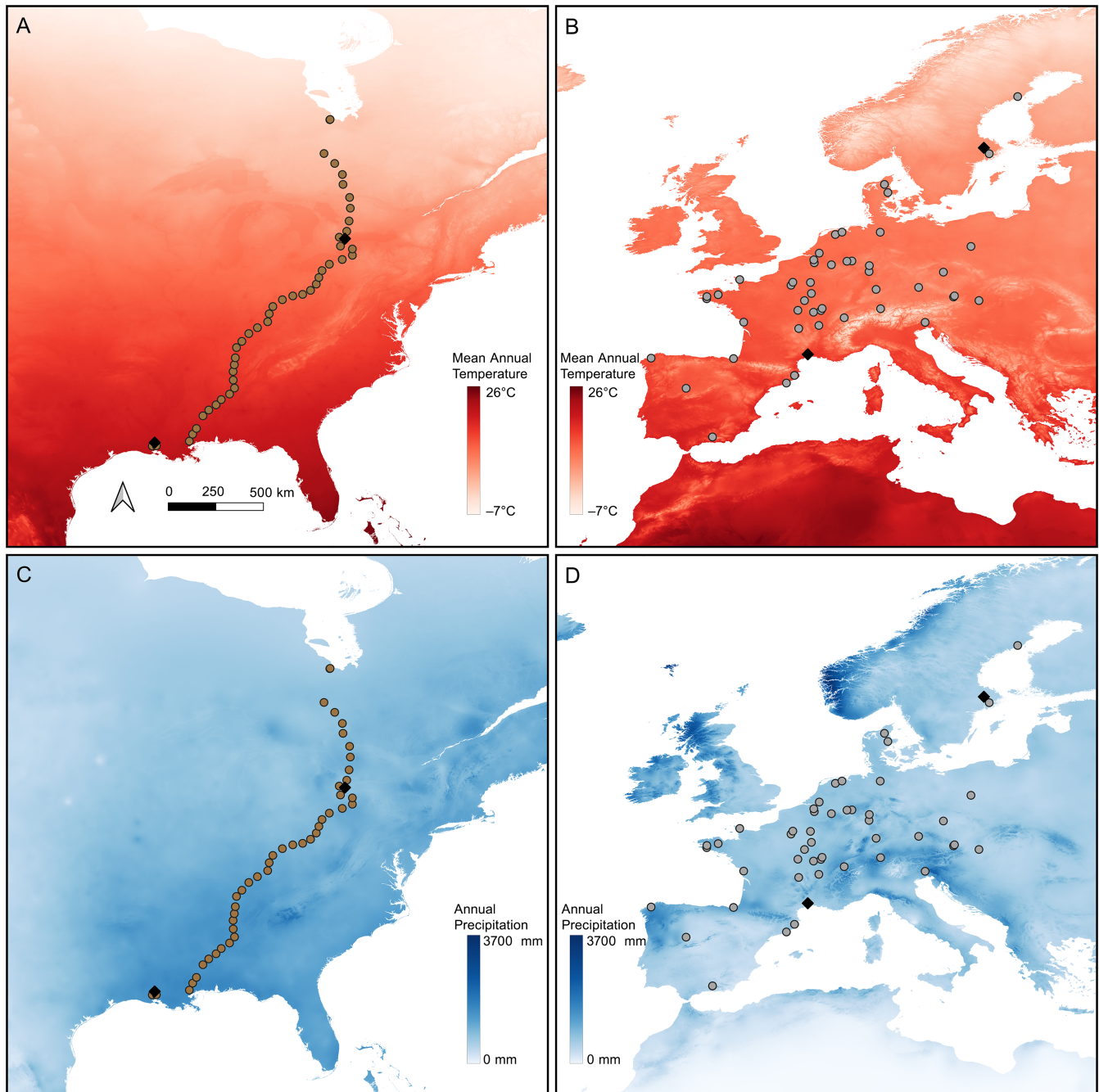


FIGURE 1 Maps showing populations of *Trifolium repens* sampled from the (A, C) introduced range (North America, brown circles) and (B, D) native range (Europe, gray circles), as well as the four common garden locations (black diamonds). Map panels (A) and (B) are set on a backdrop of color-graded historical (1970–2000) mean annual temperature (BIO1; with darker red shades representing warmer climates). Map panels (C) and (D) are set on a backdrop of color-graded historical (1970–2000) annual precipitation (BIO12; with darker blue shades representing wetter climates). Coordinates of common garden locations are 42°32′40″ N, 79°39′38″ W (Mississauga, ON, Canada), 30°18′23″ N, 92°00′33″ W (Lafayette, LA, USA), 59°49′08″ N, 17°38′49″ W (Uppsala, Sweden), and 43°38′16″ N, 3°51′43″ W (Montpellier, France). All climate raster data was extracted from WorldClim (Version 2.1, February 2023; Fick & Hijmans, 2017; O'Donnell & Ignizio, 2012).

Johnson et al. (2016). Herbivory was analyzed for the month exhibiting peak herbivory (highest average % leaf area consumed) within each garden. Multiple components of plant fitness were measured throughout the

growing seasons at each location. Survival and onset of flowering were assessed weekly during the first growing season. Mature fruits were counted and collected weekly throughout both growing seasons of the experiment, and

seeds were separated using a benchtop thresher (Precision Machine Co.; Lincoln, NE, USA) to obtain total seed set mass for each plant across both growing seasons. Photos were taken of each plant monthly during the first growing season at each common garden location and assessed for plant area using Easy Leaf Area (Version 2.0; Easlon & Bloom, 2014). The maximum plant area was then determined and the growth rate across the first growing season was calculated as $(\ln[A_{\max}] - \ln[A_1]) / (t_{\max} - t_1)$, where $A_{\max}$ represents the maximum plant area, A_1 represents the initial plant area (i.e., after 1 month of growth in the common garden), and $(t_{\max} - t_1)$ represents the number of days between photos taken for maximum plant area and initial plant area.

Calculating climatic distance

Historical (1970–2000) data for six bioclimatic variables were extracted from WorldClim (Version 2.1, February 2023) at a resolution of 30 s for each original *T. repens* population and common garden site (Fick & Hijmans, 2017; O'Donnell & Ignizio, 2012). The variables included were mean annual temperature (“BIO1”; MAT), maximum temperature of the warmest month (“BIO5”), minimum temperature of the coldest month (“BIO6”), annual precipitation (“BIO12”), precipitation of the wettest month (“BIO13”), and precipitation of the driest month (“BIO14”). The extracted values of each bioclimatic variable for each common garden location can be found in Appendix S1: Table S1. Values for all six variables were z-score standardized to allow for analysis on an equivalent scale. A principal component analysis was performed using the *princomp* function in R, with a scree plot guiding the decision to retain principal components 1, 2, and 3, which explained 65.8%, 22.1%, and 9.1% of the variation, respectively (cumulatively explaining 97% of variation; Appendix S1: Figure S1). Coordinates for each location of origin and each common garden location from all three dimensions were then extracted using the *get_pca_ind* function in the *factoextra* package (version 1.0.7; Kassambara & Mundt, 2020) and then plotted on a 3D graph (Appendix S1: Figure S2). The weighted Euclidean distance between points in 3D space was calculated using the formula:

$$\text{Distance} = \sqrt{\sum (w_i [O_{ij} - G_i]^2)}, \quad (1)$$

where w_i represents the proportion of variance explained by the i th dimension, O_{ij} represents the value of the j th population of origin in the i th dimension, and G_i represents the value of a given common garden in the i th dimension. We used this weighted Euclidean distance as a measure of the climatic distance between the location

of each population of origin and each common garden location based on our six bioclimatic variables of interest (Samis et al., 2019; Wright et al., 2018). This measure of climatic distance was then used as a predictor variable of plant fitness in our statistical analyses. Bivariate correlations among individual climatic variables and with climatic distance for each common garden are reported in Appendix S1: Table S2. We also tested average climatic distances among sampled populations within each continent transplanted into common gardens on each continent using the following model (R function *lm*):

$$\text{Climatic Distance} \sim \text{Continent of Origin (O)} + \text{Continent of Garden (G)} + \text{O : G}. \quad (2)$$

Climatic distance varied based on continent of origin ($F_{1,1854} = 93.849$, $p < 0.001$), continent of garden ($F_{1,1854} = 65.960$, $p < 0.001$), and their interaction ($F_{1,1854} = 142.491$, $p < 0.001$), with European populations transplanted into European common gardens generally experiencing the smallest climatic distances between environment of origin and the common garden environment.

Statistical analysis

All statistical procedures and analyses were conducted in R Version 4.5.0 (R Core Team, 2025) and RStudio Version 2025.05.1 (Posit Team, 2025). Seven response variables were assessed as components of fitness: (1) plant survival in the first growing season, (2) presence/absence of flowers during either growing season, (3) presence/absence of seeds during either growing season, (4) number of flower heads produced across both growing seasons, (5) seed set mass produced across both growing seasons, (6) maximum plant area in the first growing season, and (7) growth rate in the first growing season. Herbivory during the first growing season was also measured as an additional non-fitness response variable. The number of flower heads and seed set mass were both zero-inflated, which led to the use of a two-step modeling approach as described previously (Albano & Johnson, 2023). Briefly, both variables were first analyzed in binomial models to assess whether plants produced flowers or seeds. Non-zero flower head numbers and seed set masses were then subsequently analyzed as separate fitness components. All linear mixed effects models were conducted using the *glmmTMB* function in the *glmmTMB* package (version 1.1.11; Brooks et al., 2017). All test statistics and p values were calculated using the *Anova* function in the *car* package (version 3.1-3; Fox & Weisberg, 2019) with Type II sums-of-squares (SS). We fit three binary fitness measures (i.e., plant survival,

presence/absence of flowers, presence/absence of seeds) to a binomial distribution. All quantitative measures of fitness (i.e., number of flower heads, seed set mass, maximum plant area, growth rate) and herbivory were modeled using a normal distribution, with all transformations necessary to meet assumptions of normality and homogeneity of variance found in Appendix S1: Table S3. Sample sizes for each response variable and each model can be found in Appendix S1: Table S4. Slope estimates used to calculate effect sizes reported below were determined using regression lines between weighted climatic distance and the response variable. Model-predicted slope estimates and their SEs were calculated using the *emtrends* function in the *emmeans* package (version 1.11.0; Lenth, 2024).

Variation in the strength of local adaptation between common garden locations

To determine whether *T. repens* populations exhibit local adaptation and to assess the strength of local adaptation at each garden location (Q1), all response variables were analyzed using the following model structure:

$$\begin{aligned} \text{Response} \sim & \text{Continent of Origin (O)} \\ & + \text{Garden Location (L)} + \text{Climatic Distance (D)} \quad (3) \\ & + O : L + O : D + L : D + (1|\text{Population}). \end{aligned}$$

Continent of origin (North America or Europe), garden location (Mississauga, Lafayette, Uppsala, or Montpellier), and climatic distance (the combined measure of six bioclimatic variables of interest described above) were included as fixed effects, along with all two-way interactions. Population was included as a random effect to account for non-independence of replicates within each population. The strength of local adaptation refers to the magnitude of the fitness advantage for local populations compared to populations transplanted at an increasingly greater climatic distance from their origin. We therefore interpret the slope of the relationship between climatic distance and fitness as being representative of the strength of local adaptation (a negative slope of greater magnitude represents greater local adaptation strength), which can then be compared across levels for any categorical predictors that interact with climatic distance. An interaction between continent of origin and climatic distance would indicate that the strength of local adaptation varies based on whether populations originated in the native or introduced range. An interaction between garden location and climatic distance would indicate that the strength of local adaptation varies based on the common environment in

which it is being tested. Once significant interactions were identified (Table 1; Appendix S1: Table S5), all effect sizes comparing slopes to each other reported below were made using the raw data for ease of visualization and interpretation, while model-predicted slopes can be found in Table 2 and Appendix S1: Table S6.

Variation in the strength of local adaptation between ranges

To assess the strength of local adaptation between the native and introduced ranges of *T. repens* (Q2), the analyses of the previous subsection were repeated, with the continent of the common garden (North America or Europe) replacing the location of the common garden in the models:

$$\begin{aligned} \text{Response} \sim & \text{Continent of Origin (O)} \\ & + \text{Continent of Garden (G)} + \text{Climatic Distance (D)} \\ & + O : G + O : D + D : G + O : D : G + (1|\text{Population}). \end{aligned} \quad (4)$$

Interpretation of this model is similar to the model above, with an interaction between continent of origin and climatic distance indicating that the strength of local adaptation varies based on whether populations originated in the native or introduced range. An interaction between continent of garden and climatic distance would indicate that the strength of local adaptation varies based on the continent of the common garden environment in which it is being tested. In this model, we are particularly interested in a three-way interaction between continent of origin, continent of garden, and climatic distance because it would indicate that the strength of local adaptation varies based not only on the range from which populations originated but also the range in which they are being tested. In other words, this model aims to test whether locally adapted European populations maintain signatures of local adaptation in North America and vice versa. As before, once significant interactions were identified (Table 3; Appendix S1: Table S7), all effect sizes comparing slopes to each other reported below were made using the raw data for ease of visualization and interpretation, while model-predicted slopes can be found in Table 4 and Appendix S1: Table S8.

The roles of cyanogenesis and herbivory in local adaptation

The presence of cyanogenesis clines across latitude on both continents was determined separately for *Ac/ac* and *Li/li* by assessing the proportion of plants in each

TABLE 1 Results of mixed effects models assessing how continent of origin (O), common garden location (L), and climatic distance (D) affected the number of flower heads and seed set mass of *T. repens*.

Predictor	No. of flower heads			Seed set mass		
	χ^2	df	<i>p</i>	χ^2	df	<i>p</i>
Continent of origin (O)	2.270	1	0.132	1.136	1	0.286
Garden location (L)	247.413	3	< 0.001	39.550	3	< 0.001
Climatic distance (D)	13.326	1	< 0.001	14.168	1	< 0.001
O × L	1.415	3	0.702	4.059	3	0.255
O × D	0.271	1	0.602	0.873	1	0.350
L × D	12.474	3	0.006	8.553	3	0.036

Note: Significant effects ($p < 0.05$) are listed in bold.

TABLE 2 Model-predicted slope estimates and standard errors (SE) for each common garden location to describe trends in the number of flower heads and seed set mass of *T. repens* based on the interaction between garden location and climatic distance.

Garden location	No. of flower heads		Seed set mass	
	Slope	SE	Slope	SE
Lafayette	−0.597	0.168	−0.674	0.223
Mississauga	−0.169	0.170	−0.273	0.217
Montpellier	−0.037	0.252	0.053	0.316
Uppsala	0.036	0.147	0.002	0.185

population containing at least one dominant *Ac* or *Li* allele using the following model structure fit to a binomial distribution:

$$\text{Proportion } Ac \text{ or } Li \text{ Plants} \sim \text{Latitude} + \text{Continent of Origin} + \text{Latitude} : \text{Continent of Origin} + (1|\text{Population}). \quad (5)$$

An interaction between latitude and continent of origin would indicate a difference in the slope of clines in the presence of *Ac* and/or *Li* based on the range in which those clines are occurring (native European or introduced North American).

The effect of cyanotype (*AcLi*, *Acli*, *acLi*, or *acli*) on local adaptation (Q3) was assessed for all eight response variables using the following model structure:

$$\text{Response} \sim \text{Cyanotype (C)} + \text{Garden Location (L)} + \text{Climatic Distance (D)} + C:L + C:D + C:L:D + (1|\text{Population}). \quad (6)$$

An interaction between cyanotype and climatic distance would indicate that cyanogenesis or its metabolic components contribute to the strength of local adaptation in *T. repens*. A three-way interaction between cyanotype, garden location, and climatic distance would indicate

that selection for or against a given cyanotype in a particular common environment impacts the strength of local adaptation in that environment. This might be expected due to consistently observed latitudinal cyanogenesis clines across multiple types of environmental gradients in *T. repens*, which indicate selective advantages for particular cyanotypes, depending on the location in which they are found and the biotic and abiotic factors imposing selection in that location.

Bioclimatic variables as individual drivers of local adaptation

We also measured associations between the seven fitness response variables and each of the six bioclimatic variables of interest individually by determining the difference in *z*-score standardized values for each variable between the location of each population of origin and each common garden location using the following model structure:

$$\text{Response} \sim \text{Bioclimatic Variable} + \text{Bioclimatic Variable}^2 + (1|\text{Population}). \quad (7)$$

For this set of analyses, all common gardens were analyzed separately, and all quantitative (non-binomial)

response variables were z -score standardized to allow for simultaneous direct comparisons of the relative strength of linear and quadratic predictor variables. Within each garden, we were therefore able to directly compare p values across all bioclimatic variables for both linear and quadratic fits (Appendix S1: Tables S9–S12). For each response variable, we selected the most influential bioclimatic variable (i.e., the lowest p value, but only when at least one p value was significant at a level of $p < 0.1$, highlighted in red in Appendix S1: Tables S9–S12). Across all response variables in each garden, we then calculated the percentage of these most influential bioclimatic variables that were either temperature-based (BIO1, BIO5, or BIO6) or precipitation-based (BIO12, BIO13, or BIO14), and compared these percentages across the entire experiment, across latitudes, and across ranges (Q4). This approach was used to identify the most important abiotic variables driving local adaptation in general, as opposed to focusing on the results of any one particular test.

Adaptation lags in response to climate warming

We sought to investigate the presence of adaptation lags using a space-for-time substitution (Q5). We performed an additional analysis, employing a similar model structure as before but only using changes in raw values for MAT (i.e., not z -score standardized, hereafter referred to as Δ MAT) as the sole bioclimatic variable of interest.

$$\text{Response} \sim \Delta\text{MAT} + \Delta\text{MAT}^2 + (1|\text{Population}). \quad (8)$$

If the peak of the quadratic relationship in each model occurs at a Δ MAT that is approximately equal to 0°C , it would be considered evidence of local adaptation to MAT, because maximum fitness occurs in populations with MAT similar to the common garden site. If the peak of the quadratic relationship is instead shifted towards populations of warmer origins than are occurring at the common garden site (i.e., these warmer-adapted populations are outperforming the local populations), this would be considered evidence of an adaptation lag. We tested whether the shift of peaks away from a Δ MAT of 0°C and towards populations of warmer origins were significant by also fitting a similar model to above, but excluding the linear predictor:

$$\text{Response} \sim \Delta\text{MAT}^2 + (1|\text{Population}). \quad (9)$$

This additional model constrains the peak of the distribution to be centered around a Δ MAT of 0°C (approximating the outcome of local adaptation as opposed to an

adaptation lag). We then compared the models with and without the linear predictor using the *anova* and *AIC* functions in *R*, with p values < 0.05 (along with Δ AIC values < 0) indicating a significant shift of the peak.

We focused our interpretation of this analysis solely on the North American populations in the Mississauga common garden because the sampling design of North American populations along a transect and the location of the Mississauga garden within that sampled transect provide the most appropriate subset of the data to detect spatial lags. We include the results for European populations within the Uppsala common garden in Appendix S1: Table S13 for completeness, but the Uppsala common garden did not exhibit any trend in fitness based on climatic distance, precluding the possibility of an adaptation lag being detected. The Lafayette and Montpellier common gardens were excluded from this analysis because they are near the southern (warmest) extremes of our sampled populations, precluding the ability to detect adaptation lags based on Δ MAT.

RESULTS

Q1: Are *T. repens* populations locally adapted to their climate of origin and does the strength of local adaptation differ across space?

We found strong evidence of local adaptation, and the strength of local adaptation in *T. repens* varied among gardens for multiple fitness components. Climatic distance between each population of origin and the common garden locations strongly predicted all sexual fitness components and survival (Table 1; Appendix S1: Table S5). Overall, populations tended to have higher fitness with lower climatic distance, indicating that populations of more similar climatic origins to the climate of the common gardens were better adapted. The only response variables not showing evidence of local adaptation were the measures of vegetative growth used as proxies for clonal fitness (i.e., maximum plant area and growth rate). The interaction of common garden location and climatic distance affected the number of flower heads and seed set mass, but not other fitness components or herbivory (Tables 1, 2; Appendix S1: Tables S5, S6, Figure S3). In general, the strength of local adaptation, measured as the slope of the relationship between sexual fitness and climatic distance, was greater in the more southern common garden location on each continent (Figure 2). For the number of flower heads, the slope for Lafayette (-54.9) was $3.5\times$ steeper than the slope for Mississauga (-15.8) and the slope for Montpellier was highly negative (-40.0), while the slope

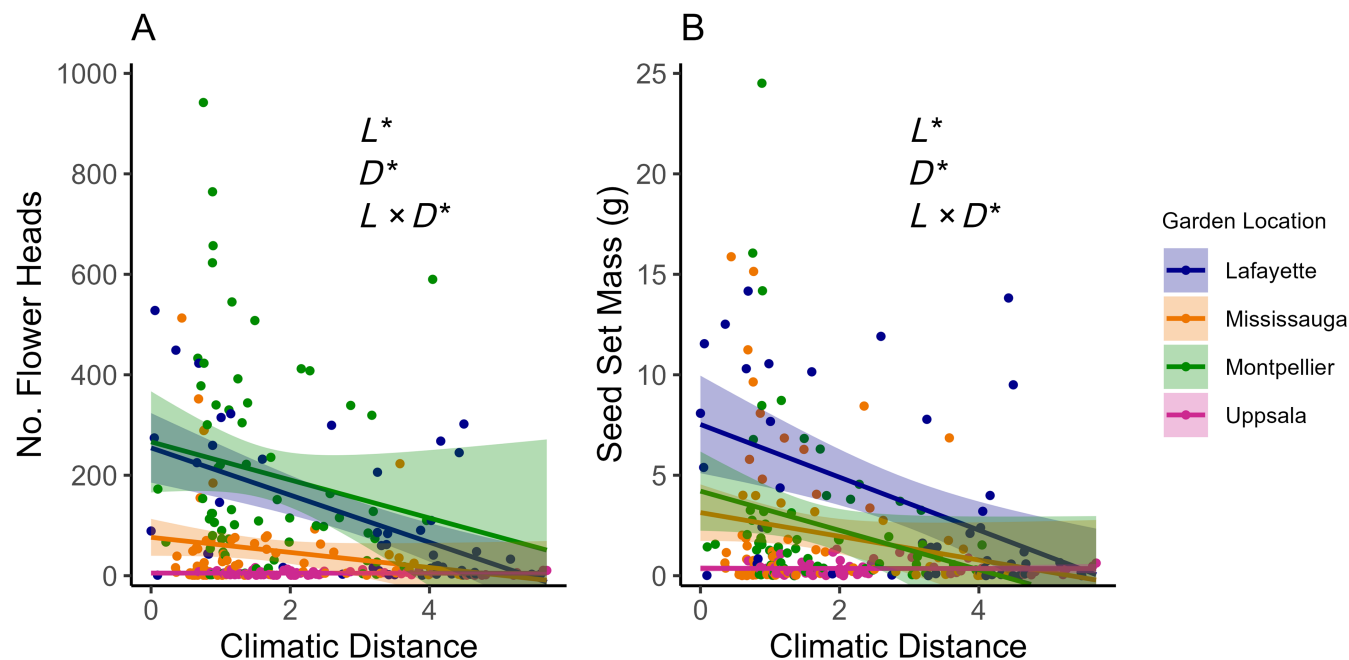


FIGURE 2 The effect of climatic distance and common garden location on fitness of *Trifolium repens*. (A) Mean number of flower heads and (B) mean seed set mass of *T. repens* populations planted in two common gardens in North America (Lafayette in navy and Mississauga in orange) and two common gardens in Europe (Montpellier in green and Uppsala in magenta). All points and trendlines displayed represent raw data in the original units for ease of visualization and interpretation. Shaded areas represent 95% CIs around each line, while asterisks represent p values for relevant fixed effects of Garden Location (L), Climatic Distance (D) and their interaction ($L \times D$) (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

for Uppsala was very slightly positive (0.1; indicating a lack of local adaptation). Similarly, for seed set mass, the slope for Lafayette (−1.4) was 2.3× steeper than the slope for Mississauga (−0.6) and the slope for Montpellier (−1.1) was 385× steeper than the slope for Uppsala (-2.9×10^{-3}). The main effect of garden location (after partitioning variance due to other main effects in the model using Type II SS, as described in the methods) affected all fitness components and herbivory, which reflects the large differences in environment between the gardens. By contrast, the main effect of continent of origin and its interactions with garden location and/or climatic distance did not affect any fitness component or herbivory (Table 1; Appendix S1: Table S5).

Q2: Does *T. repens* exhibit stronger local adaptation in its native range than in its introduced range?

We observed strong local adaptation in both the native and introduced ranges of *T. repens*, but the presence of local adaptation was dependent on where populations originated. This result is explained by a three-way interaction between continent of origin, continent of the common garden, and

climatic distance, which affected both quantitative sexual fitness components, survival, and growth rate of *T. repens* (Tables 3, 4; Appendix S1: Tables S7, S8). Specifically, populations only exhibited strong signatures of local adaptation when they were transplanted to common gardens on the same continent from which they originated (Figure 3). For number of flower heads, in common gardens located in the native range (Europe), the slope for European *T. repens* populations (−102.2) was 12× steeper than the slope for North American populations (−8.5). In common gardens located in the introduced range (North America), the slope for North American *T. repens* populations (−40.7) was 20× steeper than the slope for European populations (−2.0). Similarly, for seed set mass, in common gardens located in Europe, the slope for European *T. repens* populations (−1.4) was 24× steeper than the slope for North American populations (-5.7×10^{-2}). In common gardens located in North America, the slope for North American *T. repens* populations (−1.1) was 8.0× steeper than the slope for European populations (−0.1). The main effect of climatic distance (after partitioning variance due to main effects in the model using Type II SS) affected whether plants flowered and produced seeds, but climatic distance did not interact with the continent of origin or the continent of the common garden to affect these fitness components

(Appendix S1: Table S7, Figure S4). However, climatic distance and continent of origin did interact to affect maximum plant area and herbivory. Together across all gardens, plants from European origins that were transplanted shorter climatic distances grew larger, despite experiencing greater levels of herbivory, while size and herbivory for plants from North American origins were not significantly affected by climatic distance overall (Appendix S1: Table S7, Figure S4).

Q3: What are the roles of cyanogenesis and herbivory in driving local adaptation across space in *T. repens*?

Both *Ac/ac* and *Li/li* exhibited strong latitudinal gene frequency clines among *T. repens* populations from both continents (Appendix S1: Table S14). However, the lack of an interaction between latitude and continent of origin for both loci indicates no differences in the steepness of cyanogenesis clines between the native and introduced ranges of *T. repens* (Appendix S1: Figure S14). The main effect of cyanotype (after partitioning variance due to main effects in the model using Type II SS) affected survival, whether plants produced flowers and/or seeds, and maximum plant area, with cyanogenic plants generally exhibiting greater fitness than acyanogenic plants, regardless of the environment in which the common garden was found (Appendix S1: Table S15). However, cyanotype alone did not affect the level of herbivory occurring on *T. repens* plants, and cyanotype did not interact with either common garden location or climatic distance to affect any fitness components or herbivory (Appendix S1: Table S15).

Q4: What bioclimatic variables have the strongest impact on fitness across space in *T. repens*?

Climatic variables related to both temperature and precipitation predicted variation in fitness in *T. repens*, but the most influential predictors varied between garden locations, with patterns emerging based on latitude (Appendix S1: Tables S9–S12, red highlights). Overall, 58% of the most influential predictors were temperature-based (mean annual temperature, maximum temperature of the warmest month, or minimum temperature of the coldest month), while 42% were precipitation-based (annual precipitation, precipitation of the wettest month, or precipitation of the driest month). In the more southern gardens on each continent (Lafayette and Montpellier), 67% of the most influential

predictors were precipitation-based, while only 33% were temperature-based. Conversely, in the more northern gardens on each continent (Mississauga and Uppsala), 80% of the most influential predictors were temperature-based, while only 20% were precipitation-based. We did not find evidence that the native and introduced ranges of *T. repens* were differentially affected by temperature or precipitation variables. In the native range (Uppsala and Montpellier, combined), 57% of the most influential predictors were temperature-based, and 43% were precipitation-based. In the introduced range (Mississauga and Lafayette, combined), 58% of the most influential predictors were temperature-based, and 42% were precipitation-based.

Q5: Are there adaptation lags to climate change in *T. repens*?

North American populations of *T. repens* demonstrated evidence of an adaptation lag in the northern garden in the introduced range (Table 5). Survival, seed set mass, and maximum plant area all exhibited a significant quadratic relationship with ΔMAT between locations of origin and the Mississauga common garden (Figure 4). A quadratic relationship centered on a ΔMAT of 0°C would indicate local adaptation to the climate surrounding the common garden, while a relationship centered on a $\Delta\text{MAT} > 0^\circ\text{C}$ would represent an adaptation lag. For the proportion of plants survived per population, the peak of the quadratic relationship experienced a significant shift to a climate of origin with a MAT 3.4°C greater than Mississauga ($\chi^2_{1,4} = 4.159$, $p = 0.041$, $\Delta\text{AIC} = -2.159$) and the predicted maximum survival being 5.3% greater than at a ΔMAT of 0°C . For mean seed set mass per population, the peak of the quadratic relationship occurred at a MAT 2.0°C greater than Mississauga, with the predicted maximum seed set mass being 5.2% greater than at a ΔMAT of 0°C . However, this shift was not statistically significant ($\chi^2_{1,5} = 0.640$, $p = 0.423$, $\Delta\text{AIC} = 1.360$). For mean maximum plant area per population, the peak of the quadratic relationship experienced a shift to a climate of origin with a MAT 2.1°C ($\chi^2_{1,5} = 3.721$, $p = 0.054$, $\Delta\text{AIC} = -1.721$) greater than Mississauga, with the predicted maximum plant area being 3.9% greater than at a ΔMAT of 0°C . Along our transect of sampled populations, a shift in the peak of the quadratic relationship towards populations 2.0 – 3.4°C warmer than the Mississauga common garden represents a 1.9 – 3.3° shift in latitude towards the equator. Additionally, whether plants flowered or produced seeds increased linearly with ΔMAT , indicating plants of warmer origins were more likely to flower and produce seeds (Figure 4),

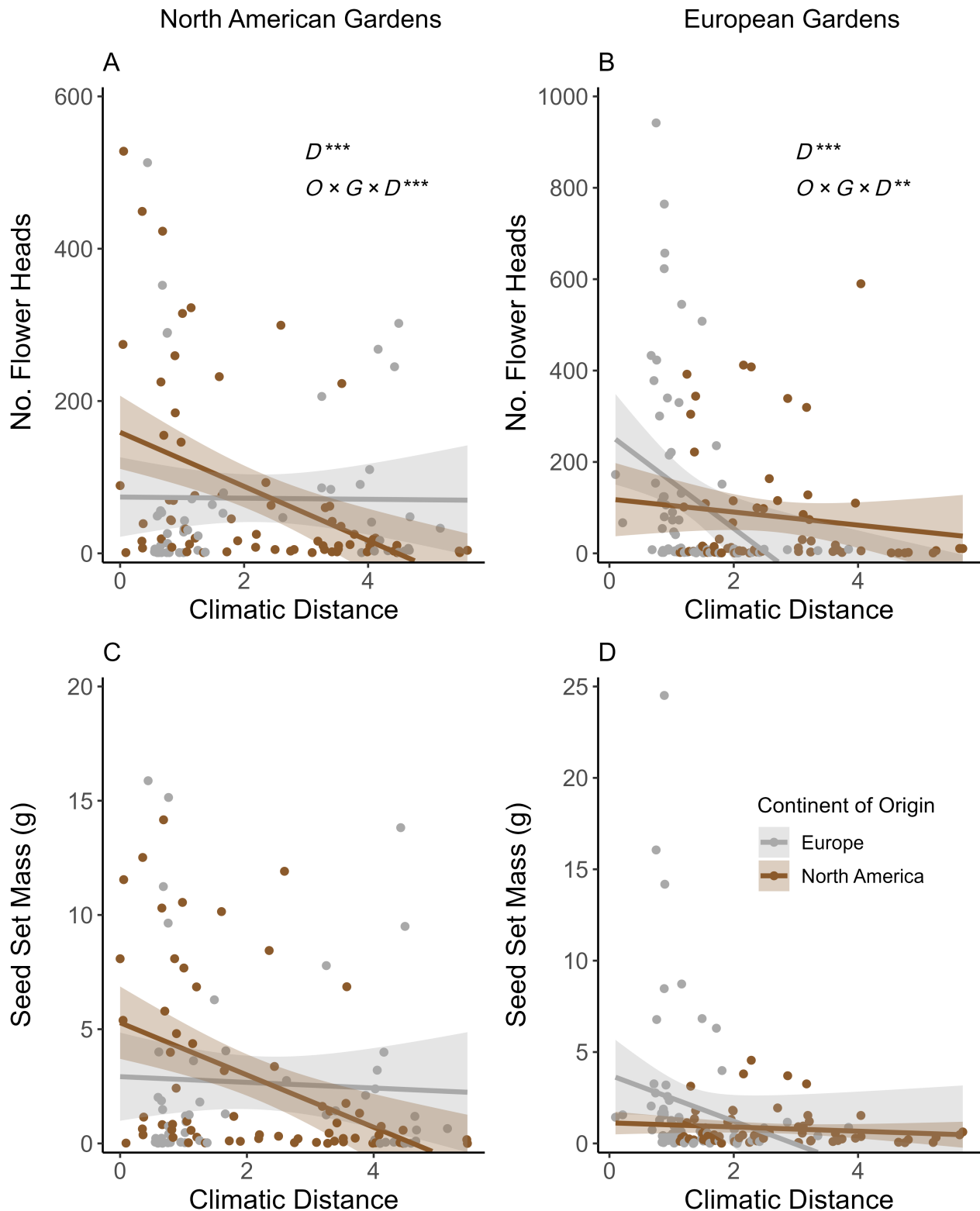


FIGURE 3 The effect of climatic distance on the mean number of flower heads produced by *Trifolium repens* populations originating from its introduced (North America, brown) and native (Europe, gray) ranges and transplanted into common gardens in the (A) introduced and (B) native ranges. The effect of climatic distance on the mean seed set mass produced by *T. repens* populations originating from its introduced and native ranges and transplanted into common gardens in the (C) introduced and (D) native ranges. All points and trendlines displayed represent raw data in the original units for ease of visualization and interpretation. Shaded areas represent 95% CIs around each line, while asterisks represent p values for relevant fixed effects of Climatic Distance (D) and its interaction with Continent of Origin (O) and Continent of Garden (G) (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

TABLE 3 Results of mixed effects models assessing how continent of origin (O), continent of garden (G), and climatic distance (D) affected the number of flower heads and seed set mass of *T. repens*.

Predictor	No. of flower heads		Seed set mass	
	χ^2	<i>p</i>	χ^2	<i>p</i>
Continent of origin (O)	0.514	0.474	0.028	0.867
Continent of garden (G)	0.471	0.493	0.037	0.847
Climatic distance (D)	14.310	< 0.001	16.190	< 0.001
O × G	0.957	0.328	0.481	0.488
O × D	1.214	0.271	2.991	0.084
G × D	1.412	0.235	0.258	0.612
O × G × D	19.603	< 0.001	9.728	0.002

Note: Significant effects ($p < 0.05$) are listed in bold. All predictors are assessed using one degree of freedom.

TABLE 4 Model-predicted slope estimates and standard errors (SE) for each common garden location to describe trends in the number of flower heads and seed set mass of *T. repens* for each combination within the interaction of continent of origin and continent of common garden.

Cont. of origin/cont. of garden	No. of flower heads		Seed set mass	
	Slope	SE	Slope	SE
North America/North America	−0.482	0.125	−0.631	0.139
North America/Europe	−0.237	0.162	−0.179	0.169
Europe/North America	0.097	0.139	−0.006	0.151
Europe/Europe	−1.351	0.301	−0.826	0.313

TABLE 5 Results of mixed effects models assessing how change in mean annual temperature (Δ MAT) affected the fitness of North American *T. repens* populations in the Mississauga common garden (linear and quadratic fits).

Predictor	Plants producing flowers		Plants producing seeds		No. of flower heads		Seed set mass		Survival		Maximum plant area		Growth rate	
	χ^2	<i>p</i>	χ^2	<i>p</i>	χ^2	<i>p</i>	χ^2	<i>p</i>	χ^2	<i>p</i>	χ^2	<i>p</i>	χ^2	<i>p</i>
Δ MAT	7.425	0.006	5.397	0.020	0.009	0.924	0.644	0.422	3.758	0.053	3.799	0.051	0.002	0.962
Δ MAT ²	8.077	0.004	7.609	0.006	2.160	0.142	8.542	0.003	4.313	0.038	6.741	0.009	1.808	0.179

Note: Significant effects ($p < 0.05$) are listed in bold. All predictors are assessed using one degree of freedom.

while the number of flower heads and growth rate did not exhibit any significant linear or quadratic trends with Δ MAT (Table 5).

DISCUSSION

The aim of our study was to provide insight into the potential for rapid local adaptation in introduced plant populations in response to novel environments across space and time. The results of our transcontinental experiment in the native (Europe) and introduced

(North America) ranges of *T. repens* lead to two main conclusions: (1) rapid local adaptation to climatic variation across space has contributed to the ecological success of *T. repens* in its introduced range; and (2) climate warming has resulted in some evidence of a weakening of local adaptation over time in the introduced range of *T. repens*. Our conclusions are based on five experimental results. First, *T. repens* exhibited local adaptation to climate overall, with greater fitness advantages for local populations at lower latitudes (Q1). Second, in European common gardens, only native European populations exhibited local adaptation, while in North

American common gardens, only introduced North American populations exhibited local adaptation (Q2). Third, cyanogenesis did not differentially affect the fitness of *T. repens* in different environments (Q3). Fourth, both temperature- and precipitation-based bioclimatic variables predicted fitness, but temperature was a stronger determinant of fitness at higher latitudes, while precipitation was a stronger determinant of fitness at lower latitudes (Q4). Fifth, there is some evidence consistent with an adaptation lag in response to climate warming, but the signature was weak, and this pattern was only clear in the northern common garden of the introduced range (Q5). Below, we discuss our results in the context of the role of rapid local adaptation in the success of non-native species and how introduced populations may (or may not) be able to adapt to future climate change.

Strength of local adaptation is dependent on latitude

Spatial variation in biotic and abiotic factors across a species' range could affect its ability to adapt to different environments within that range. We found that *T. repens* exhibited significant local adaptation to the abiotic environment based on a measure of climatic distance that encompassed mean annual temperature and precipitation along with annual temperature and precipitation extremes. However, local adaptation differed in strength depending on the latitude of common gardens. Specifically, local adaptation was stronger at lower latitudes in both the native and introduced ranges of *T. repens* when fitness was measured as sexual reproductive success (i.e., number of flower heads and seed set mass). Our results are consistent with transplant experiments in other plant species, which found sexual fitness advantages for local versus foreign plants to be more pronounced at lower latitudes (Colautti & Barrett, 2013; Tor  ng et al., 2015), likely at least partially due to life history trade-offs across large latitudinal ranges (Bastias et al., 2024; Li et al., 2015). In fact, life history trade-offs in *T. repens* contribute to local adaptation across divergent environments, with southern populations investing more into sexual reproduction and northern populations investing more into vegetative growth (Wright et al., 2022). If warmer climates are generally more conducive to sexual reproduction than cooler climates, this may explain why sexual fitness advantages for local populations were more pronounced in Lafayette and Montpellier in our study, while climatic distance had more limited effects on vegetative growth.

Local adaptation in native and introduced ranges

Non-native species could become more widespread and persistent by rapidly adapting to novel environments. For example, latitudinal cyanogenesis clines in introduced North American *T. repens* populations formed due to rapid local adaptation and sorting of existing genotypes post-introduction (Kooyers & Olsen, 2012, 2013; Olsen et al., 2021). We found that populations of *T. repens* only exhibited strong signatures of local adaptation to climate when transplanted into common gardens within the same range as they originated from. The observed strong local fitness advantages for North American populations within the North American gardens are therefore indicative of rapid local adaptation of *T. repens* within the past 400 years since its introduction. Strong selection on introduced populations provides opportunity for rapid local adaptation in non-native species on even shorter time-scales than exhibited by *T. repens* (Colautti & Lau, 2015; Sim  n-Porcar et al., 2021). A meta-analysis by Oduor et al. (2016) found that non-native populations exhibited local adaptation just as frequently as native populations, and that the strength of local adaptation was not affected by time since introduction (ranging from as little as 20 to as many as 500 years). More specifically, previous studies on invasion in *Hypericum canariense* (Dlugosch & Parker, 2008a), *Eschscholzia californica* (Leger & Rice, 2007), and *Lythrum salicaria* (Colautti & Barrett, 2013) indicate local adaptation in introduced ranges within 50, 150, and 200 years, respectively. Taken together, our results strengthen the mounting evidence that rapid local adaptation to climate plays an important role in the ecological success of non-native species in their new range.

Interestingly, *T. repens* populations from the introduced range did not exhibit signatures of local adaptation to similar climates when transplanted back into their native range. Similarly, native populations did not maintain their strong signatures of local adaptation to climate when transplanted into the introduced range. One important caveat is the overall narrower range of climatic distances among European populations (particularly within European common gardens), despite our widespread sampling (Appendix S1: Figure S2), which could have limited our ability to compare the strength of local adaptation between ranges (Lucas et al., 2024). However, these climatic differences between ranges are important aspects of the biology of the plants that could affect selection in their introduced range, while the standardized measure of climatic distance helps account for some of the environmental discrepancy between ranges. Additionally, while our common gardens are robust to

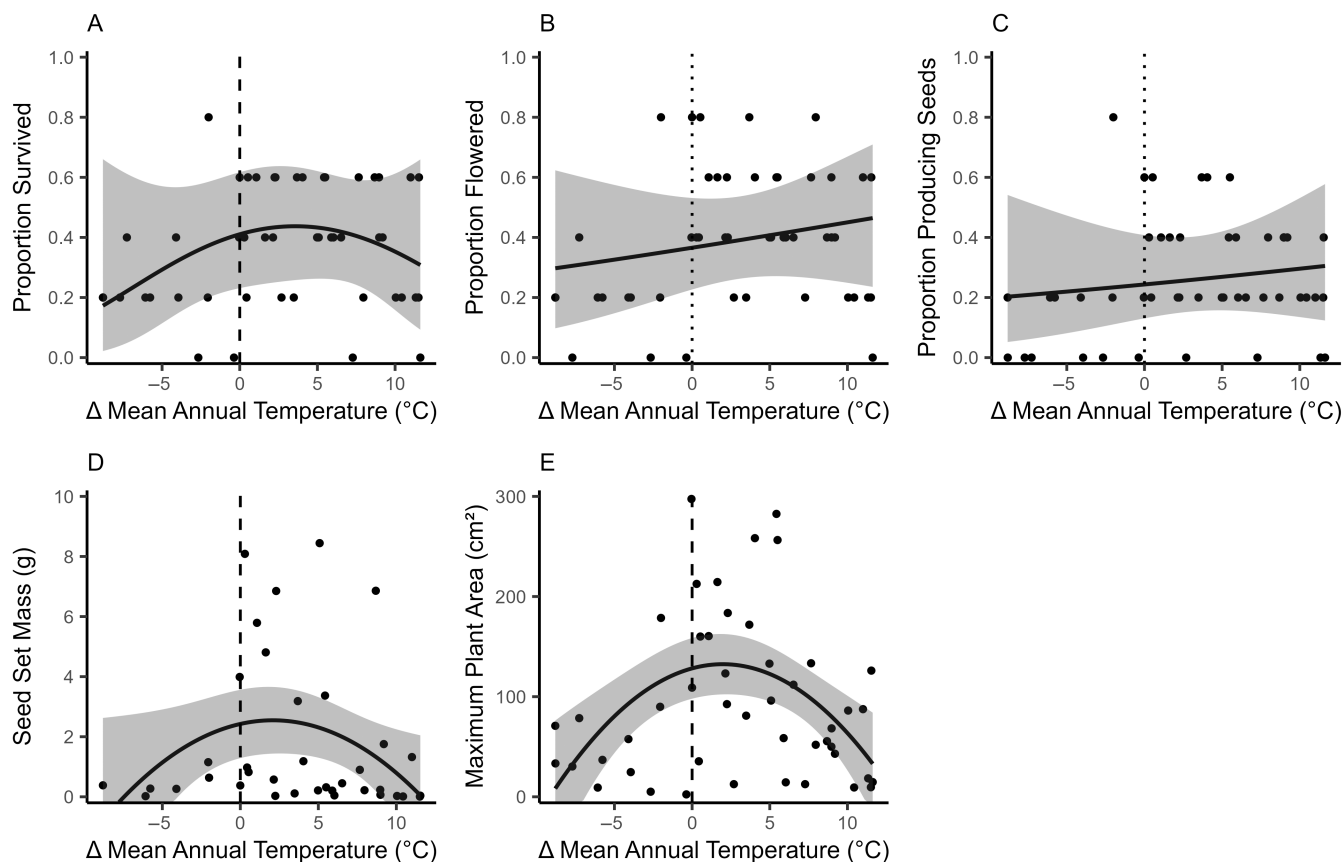


FIGURE 4 The effect of the change in mean annual temperature (Δ MAT) on the fitness of North American populations of *Trifolium repens* grown in the Mississauga common garden. (A) Proportion of plants that survived per population, (B) proportion of plants that produced flowers per population, (C) proportion of plants that produced seeds per population, (D) mean seed set mass per population, and (E) mean maximum plant area per population all exhibited at least qualitative evidence of an adaptation lag in response to climate warming, indicated by a positive linear trend or a shift in the peak of the quadratic curve towards populations with warmer climates of origin (denoted as x -axis values > 0). All points and trendlines displayed represent raw data in the original units for ease of visualization and interpretation. Shaded areas represent 95% CIs around each line.

detecting local adaptation caused by genotype $\times$ environment interactions (Bock et al., 2018), we also acknowledge that phenotypic and/or transgenerational plasticity (whose determination was beyond the scope of this study) may also be important for successful establishment or spread of non-native species (de Villemereuil et al., 2018; Hulme, 2008; Valladares et al., 2014). Despite these considerations, our findings demonstrate a clear cost of rapid adaptation, with local fitness advantages not extending across continents, even when accounting for a comprehensive measure of climatic distance between source environments and common garden environments on both continents (Pal et al., 2020). Explanations for this lack of maintenance of local adaptation across continents could be due to unmeasured biotic or abiotic aspects of the environment.

Abiotic explanations for the lack of local adaptation across continents could involve shifts in photoperiod or precipitation regimes between Europe and North

America due to similar climates on the two continents being associated with different latitudes. In general, European environments have longer day lengths in summer and shorter day lengths in winter than those of similar climates in North America, with differences being more pronounced at progressively higher latitudes. This photoperiodic shift between the continents could have affected the initiation of flowering and/or the timing of peak flowering for populations that were transplanted across continents, leading to differences in the strength of local adaptation. Halbritter et al. (2015) found asymmetries in the strength of local adaptation in reciprocally transplanted high-elevation and high-latitude *Plantago lanceolata* and *P. major* populations with similar climates of origin, indicating a potential role of photoperiod in local adaptation. Additionally, populations of an obligate long-day plant, *Mimulus guttatus*, exhibit clinal variation in the critical photoperiod required to initiate flowering, similarly indicating its potential role in

driving local adaptation (Kooyers et al., 2015). Saikkonen et al. (2012) speculated that the transition to less extreme yearly variation in photoperiod at lower latitudes might facilitate greater ecological success of European species introduced to North America than vice versa, which could be especially true of a facultative long-day plant like *T. repens* (Beatty, 1959). However, our results do not provide evidence of increased success of European populations in North American gardens compared to that of North American populations in European gardens, indicating that other environmental factors are likely at play. Additionally, although we accounted for mean annual precipitation and yearly precipitation extremes in our study, these variables do not fully capture the exact timing and volume of precipitation, and the resulting hydrological environment experienced by *T. repens* plants in Europe versus North America. For example, in the Montpellier garden in southern Europe, plants will experience a Mediterranean climate with a precipitation regime that is both dry and seasonally variable, while in the Lafayette garden in southern North America, plants will experience a subtropical environment that is wetter with more limited seasonal variation. This shift in precipitation availability has caused introduced *T. repens* populations to evolve greater drought avoidance as opposed to the more desiccation-tolerant strategy observed in Mediterranean populations (Hendrickson et al., 2025).

While our experiment offers strong tests of local adaptation to both biotic and abiotic factors in the native and introduced ranges of a cosmopolitan plant, one limitation of this study is an inability to determine the relative strength of soil conditions compared with climate as drivers of local adaptation. Planting *T. repens* individuals from each population along with the natural soil and microbial communities from their location of origin was infeasible considering our study design and governmental regulations on importation of soil and microbes. Instead, the best alternative was to maintain consistent treatment of all individuals in all gardens, with *T. repens* individuals planted into the natural soil at each common garden site. It is important to recognize that local edaphic factors have the potential to provide fitness advantages in *T. repens*, as has been demonstrated in other systems (Macel et al., 2007; Murray-Stoker & Johnson, 2024). Despite this, signatures of local adaptation to climate were still evident across multiple fitness components, strengthening our confidence in the observed patterns of rapid adaptation across ranges. However, a future study exploring the roles of edaphic and other abiotic factors could provide more comprehensive insight into how introduced plant populations adapt to new ranges.

Cyanogenesis did not affect local adaptation in either the native or introduced range

The biotic environment could also partially explain the lack of local adaptation across continents based on climatic distance (Hargreaves et al., 2020). In line with classic hypotheses in invasion biology, herbivory could exhibit substantial differences across environments based on expected shifts in herbivore communities between ranges, which could facilitate success of a plant species post-introduction (Blossey & Nötzold, 1995; Coley et al., 1985; Keane & Crawley, 2002). If herbivory was likely to explain our results, we might expect to observe signatures of local adaptation based on the presence of chemical defenses in *T. repens* plants. Among our sampled populations, we found rapid evolution of latitudinal clines in the presence of both *Ac* and *Li* alleles in the introduced range of *T. repens*, based on their similar strength to clines in the native range, demonstrating potential for involvement of cyanogenesis in local adaptation. However, our experimental results indicate that HCN did not confer higher fitness to local plants compared to foreign plants in any garden. In fact, local plants exhibited signatures of local adaptation despite experiencing similar or greater levels of herbivory than populations transplanted from further climatic distances. Additionally, experimental plants in the introduced range exhibited greater herbivory than plants in the native range, which is contrary to the prediction that *T. repens* was released from natural enemies in its native range (Keane & Crawley, 2002). Previous studies on local adaptation in *T. repens* have similarly found that HCN only had marginal, if any, effects on fitness (Wright et al., 2018, 2022) or have found weaker clines in the introduced range than the native range (Daday, 1954a, 1958; Innes et al., 2022). Taken together, our study contrasts classic invasion hypotheses, providing little evidence of a role of biotic factors (cyanogenesis or herbivory) in driving local adaptation of *T. repens* to its introduced range.

Our findings also differ from previous work that has speculated on the involvement of trade-offs in the costs and benefits of cyanogenesis or its metabolic components in adaptation to contrasting environments. For example, exacerbated ecological or allocation costs of producing HCN or its metabolic components in cooler climates could pair with exacerbated benefits of cyanogenesis in warmer climates with greater herbivore pressure (Daday, 1954a, 1965; Fadoul et al., 2023; Kooyers et al., 2018). In our study, cyanogenic plants often demonstrated a fitness advantage (i.e., benefits outweighing costs), even in cooler environments like Mississauga.

However, despite these fitness advantages, we did not find evidence for HCN as a deterrent to herbivory in *T. repens*, indicating that greater prevalence of cyanogenesis in warmer climates is likely not based on an increased ability to defend against greater herbivore pressure. Previous work instead indicates the potential for cyanogenic glucosides and/or linamarase, individually, to be advantageous in periods of drought through the storage and recycling of nitrogen (Kooyers et al., 2014; Machingura et al., 2016; Møller, 2010). However, we also do not find evidence of the metabolic components of cyanogenesis affecting the fitness of *T. repens* or its adaptation to local climates. It is important to highlight that we were unable to harvest enough tissue to determine the cyanotype of most experimental plants due to mortality and/or concerns over the effects of tissue harvest on plant fitness, so the strength of our conclusions with respect to the role of cyanogenesis in adaptation is limited by low sample sizes. Additionally, we do not account for the potential involvement of other biotic interactions, such as belowground herbivory, soil biota, or competition, in selection and local adaptation in *T. repens* (Blossey & Nötzold, 1995; Mitchell et al., 2006; Murray-Stoker & Johnson, 2024; Pal et al., 2020; Sheng et al., 2022). Further field studies are needed to more explicitly test a diverse array of biotic factors as drivers of local adaptation to climate across large environmental gradients, including direct tests of the costs and benefits of the most prominent defense mechanism in *T. repens*.

Spatial variation in environmental drivers of local adaptation

While *T. repens* exhibits local adaptation in its native and introduced ranges based on an overall measure of climatic distance, populations may vary spatially in the individual bioclimatic variables responsible for local adaptation. Our results across both ranges demonstrate that at higher latitudes, where local adaptation is weaker, temperature-based variables are stronger determinants of fitness, while at lower latitudes, where local adaptation is stronger, precipitation-based variables are stronger determinants of fitness. Therefore, it is possible that *T. repens* adapts more readily to local precipitation regimes than local temperature regimes, resulting in stronger signatures of local adaptation in lower latitude common gardens in both the native and introduced ranges. This result is consistent with Siepielski et al. (2017), who found both precipitation and potential evapotranspiration to be stronger predictors of selection than temperature across numerous taxa. Alternatively, our results could be affected by the specific environments chosen for

the common gardens. In our study, the climates in Lafayette and Montpellier are among the warmest that occur in the range of *T. repens*, while representing two opposing extremes in terms of precipitation regime, and therefore, could have been especially difficult to tolerate for populations transplanted from cooler climates with less extreme precipitation regimes. In turn, Uppsala was somewhat isolated, both geographically and climatically, from the sampled *T. repens* populations, which could have affected our ability to detect local adaptation (or maladaptation) in that environment. Lastly, it is possible that recent climate warming may have already disrupted the ability of local populations to outperform foreign populations, thereby impacting the strength of local adaptation in certain locations.

Evidence of adaptation lags in response to climate warming

The ability of populations to adapt to recent rapid climate warming over time is an important mechanism by which species can mitigate the risk of local extinction and other downstream ecological consequences. Previous studies have demonstrated the incidence of adaptation lags, whereby populations originating from warmer climates would outperform populations with very similar climatic origins to the common garden location (Aguirre-Liguori et al., 2019; Anderson & Wadgymar, 2020; Wilczek et al., 2014). These studies indicate the potential for a disruption in local adaptation, and even the possibility of maladaptation, due to climate warming that is occurring faster than local populations are able to adapt.

In our northern common garden in North America (Mississauga), populations with warmer origins (2–3°C in MAT) outperformed local populations in survival and biomass production (i.e., a proxy for vegetative reproduction) but not sexual fitness components. This finding may signal the presence of an adaptation lag and may also provide a partial explanation for the observed weaker local adaptation in Mississauga compared with Lafayette. However, the generalizability of our results should be considered carefully for a few reasons. First, contrary to our expectations, we provide no evidence of either local adaptation or maladaptation in Uppsala. In a large-scale common garden experiment with European populations of *Arabidopsis thaliana*, Wilczek et al. (2014) found that adaptation lags were more pronounced in higher latitude common garden locations, which was not the case in our study. It would be pertinent to be able to test for spatial lags in the native range of *T. repens* to which we could directly compare spatial lags in the introduced range. Second, we tested for the presence of adaptation lags

based solely on MAT, and Mississauga was the common garden location where MAT most consistently affected plant fitness compared to other bioclimatic variables. Fitness in lower latitude common gardens was instead more heavily influenced by precipitation-based variables, which may not be changing as consistently or directionally as temperature-based variables due to anthropogenic activity but could still play a significant role in maladaptation in response to changing environments (Nagy et al., 2024). It is therefore possible that adaptation lags to changing environments may not occur consistently across different geographic locations, based on the specific biotic and abiotic factors that most strongly impact local adaptation in those environments. Third, the MAT at the location of the Mississauga common garden in 2020 and 2021 was 1.6°C and 2.0°C warmer, respectively, than the average from the historical time period (1970–2000) used to calculate climatic distances in this study (data extracted from ClimateNA; Wang et al., 2016). Therefore, it may not be unexpected to see populations from warmer origins outperform local populations. The adaptation lag for survival does extend to a peak 3.4°C warmer than Mississauga, and the proportion of plants producing flowers and/or seeds increased linearly with Δ MAT, which still provides evidence for warm-adapted populations having an advantage that exceeds expectations based on the MAT during the years of the experiment. However, across our entire study, we are unable to make broad conclusions about future vulnerability of populations based on the rate of environmental change exceeding the rate of adaptation. Instead, we conclude that the fine-tuned nature of local adaptation (in terms of variation in strength, prominent driving factors, and the potential for maladaptation) to climate in *T. repens* highlights the multifaceted nature of rapid evolution in response to changing environments.

Conclusions

Our study provides insight into the ability of introduced populations of a widespread plant to adapt to novel environments across space and over time. Signatures of local adaptation were only evident when populations were transplanted within the same range as they originated from, indicating both strong and rapid local adaptation to the introduced range of *T. repens* in less than 400 years since introduction. Interestingly, the strength of local adaptation and the most important environmental predictors of plant fitness differed between the north and south of both the native and introduced ranges. Specifically, temperature was the most important driver of local adaptation in northern gardens, whereas

precipitation was most important in southern gardens. This result indicates that there may be predictable drivers of local adaptation of non-native populations to novel environments, and that these drivers may vary along broad latitudinal gradients. Despite evidence of rapid adaptation to changing environments across space, the northern garden in the introduced range also provided some evidence of local maladaptation compared to populations with warmer origins, which is characteristic of an adaptation lag. Therefore, while local adaptation has occurred quickly in introduced populations of *T. repens* across space, a lack of rapid adaptation to environmental change over time could eventually result in severe erosion of fitness advantages for local populations in the future. Continuing to further our understanding of the ability of native and introduced populations to rapidly adapt to anthropogenic environmental change in space and time is vital to the integrity and maintenance of biodiversity in the face of future shifts in environment.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Albano et al., 2025a) are available in Dryad at (<https://doi.org/10.5061/dryad.bk3j9kdqk>). Code (Albano et al., 2025b) is available in Zenodo at (<https://doi.org/10.5281/zenodo.17056843>).

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REFERENCES

- Adams, J. M., W. Fang, R. M. Callaway, D. Cipollini, E. Newell, and Transatlantic Acer platanoides Invasion Network (TRAIN). 2009. "A Cross-Continental Test of the Enemy Release Hypothesis: Leaf Herbivory on *Acer platanoides* (L.) Is Three Times lower in North America than in its Native Europe." *Biological Invasions* 11: 1005–16.
- Agrawal, A. A. 2007. "Macroevolution of Plant Defense Strategies." *Trends in Ecology & Evolution* 22: 103–9.
- Agrawal, A. A., A. P. Hastings, M. T. J. Johnson, J. L. Maron, and J.-P. Salminen. 2012. "Insect Herbivores Drive Real-Time Ecological and Evolutionary Change in Plant Populations." *Science* 338: 113–16.
- Agrawal, A. A., and P. M. Kotanen. 2003. "Herbivores and the Success of Exotic Plants: A Phylogenetically Controlled Experiment." *Ecology Letters* 6: 712–15.
- Aguiar-Liguori, J. A., S. Ramírez-Barahona, P. Tiffin, and L. E. Eguarte. 2019. "Climate Change Is Predicted to Disrupt Patterns of Local Adaptation in Wild and Cultivated Maize." *Proceedings of the Royal Society B: Biological Sciences* 286: 20190486.
- Albano, L. J., and M. T. J. Johnson. 2023. "Interactions between Environmental Factors Drive Selection on Cyanogenesis in *Trifolium repens*." *Oikos* 2023: e09629.
- Albano, L., C. Bastias, A. Estarague, B. Hendrickson, S. Innes, N. King, C. Patterson, et al. 2025a. "A Transcontinental Experiment Elucidates (Mal)Adaptation of a Cosmopolitan Plant to Climate in Space and Time [Dataset]." Dryad. <https://doi.org/10.5061/dryad.bk3j9kdqk>.
- Albano, L., C. Bastias, A. Estarague, B. Hendrickson, S. Innes, N. King, C. Patterson, et al. 2025b. "A Transcontinental Experiment Elucidates (Mal)Adaptation of a Cosmopolitan Plant to Climate in Space and Time." Zenodo. <https://doi.org/10.5281/zenodo.17056843>.
- Allendorf, F. W., and L. L. Lundquist. 2003. "Introduction: Population Biology, Evolution, and Control of Invasive Species." *Conservation Biology* 17: 24–30.
- Anderson, J. T., and B.-H. Song. 2020. "Plant Adaptation to Climate Change—Where Are we?" *Journal of Systematics and Evolution* 58: 533–545.
- Anderson, J. T., and S. M. Wadgymar. 2020. "Climate Change Disrupts Local Adaptation and Favours Upslope Migration." *Ecology Letters* 23: 181–192.
- Angseesing, J. P. A. 1974. "Selective Eating of the Acyanogenic Form of *Trifolium repens*." *Heredity* 32: 73–83.
- Anstett, D. N., J. R. Ahern, M. T. J. Johnson, and J.-P. Salminen. 2018. "Testing for Latitudinal Gradients in Defense at the Macroevolutionary Scale." *Evolution* 72: 2129–43.
- Atkins, K. E., and J. M. J. Travis. 2010. "Local Adaptation and the Evolution of Species' Ranges under Climate Change." *Journal of Theoretical Biology* 266: 449–457.
- Atwood, S. S., and J. T. Sullivan. 1943. "Inheritance of a Cyanogenic Glucoside and its Hydrolyzing Enzyme in *Trifolium repens*." *Journal of Heredity* 34: 311–320.
- Baskett, C. A., and D. W. Schemske. 2018. "Latitudinal Patterns of Herbivore Pressure in a Temperature Herb Support the Biotic Interactions Hypothesis." *Ecology Letters* 21: 578–587.
- Bastias, C. C., A. Estarague, D. Vile, E. Gaignon, C.-R. Lee, M. Exposito-Alonso, C. Violle, and F. Vasseur. 2024. "Ecological Trade-Offs Drive Phenotypic and Genetic Differentiation of *Arabidopsis thaliana* in Europe." *Nature Communications* 15: 5185.
- Battlay, P., B. T. Hendrickson, J. I. Mendez-Reneau, J. S. Santangelo, L. J. Albano, J. Wilson, A. E. Caizergues, et al. 2025. "Haploblocks Contribute to Parallel Climate Adaptation Following Global Invasion of a Cosmopolitan Plant." *Nature Ecology & Evolution* 9: 1441–55.
- Beatty, D. W. 1959. *Effect of Photoperiod and Temperature on Flowering of White Clover (Trifolium repens)*. Ames, IA: Iowa State University.
- Bidart-Bouzat, M. G., and A. Imeh-Nathaniel. 2008. "Global Change Effects on Plant Chemical Defenses against Insect Herbivores." *Journal of Integrative Plant Biology* 50: 1339–54.
- Blossey, B., and R. Nötzold. 1995. "Evolution of Increased Competitive Ability in Invasive Nonindigenous Plants: A Hypothesis." *Journal of Ecology* 83: 887–89.
- Bock, D. G., M. B. Kantar, C. Caseys, R. Matthey-Dorey, and L. H. Rieseberg. 2018. "Evolution of Invasiveness by Genetic Accommodation." *Nature Ecology & Evolution* 2: 991–99.
- Bontrager, M., C. D. Muir, C. Mahony, D. E. Gamble, R. M. Germain, A. L. Hargreaves, E. J. Kleynhans, K. A. Thompson, and A. L. Angert. 2020. "Climate Warming Weakens Local Adaptation." *bioRxiv*. <https://doi.org/10.1101/2020.11.01.364349>
- Brighton, F., and M. T. Horne. 1977. "Influence of Temperature on Cyanogenic Polymorphisms." *Nature* 265: 437–38.
- Brondizio, E. S., J. Settele, S. Díaz, and H. T. Ngo. 2019. *Global Assessment Report on the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services*. Bonn, Germany: IPBES Secretariat.
- Brooks, M. E., K. Kristensen, K. J. van Benthem, A. Magnusson, C. W. Berg, A. Nielsen, H. J. Skaug, M. Mächler, and B. M. Bolker. 2017. "glmmTMB Balances Speed and Flexibility among Packages for Zero-Inflated Generalized Linear Mixed Modeling." *The R Journal* 9: 378–400.
- Burdon, J. J. 1983. "*Trifolium repens* L." *Journal of Ecology* 71: 307–330.
- Burns, J. E., T.-L. Ashman, J. A. Steets, A. Harmon-Threatt, and T. M. Knight. 2011. "A Phylogenetically Controlled Analysis of the Roles of Reproductive Traits in Plant Invasions." *Oecologia* 166: 1009–17.
- Byers, J. E. 2002. "Impact of Non-Indigenous Species on Natives Enhanced by Anthropogenic Alteration of Selection Regimes." *Oikos* 97: 449–458.

- Callaway, R. M., and W. M. Ridenour. 2004. "Novel Weapons: Invasive Success and the Evolution of Increased Competitive Ability." *Frontiers in Ecology and the Environment* 2: 436–443.
- Callaway, R. M., J. E. Lucero, J. L. Hierro, and C. J. Lortie. 2022. "The EICA Is Dead? Long Live the EICA!" *Ecology Letters* 25: 2289–2302.
- Capblancq, T., M. C. Fitzpatrick, R. A. Bay, M. Exposito-Alonso, and S. R. Keller. 2020. "Genomic Prediction of (Mal)Adaptation across Current and Future Climatic Landscapes." *Annual Review of Ecology, Evolution, and Systematics* 51: 245–269.
- Colautti, R. I., and J. A. Lau. 2015. "Contemporary Evolution during Invasion: Evidence for Differentiation, Natural Selection, and Local Adaptation." *Molecular Ecology* 24: 1999–2017.
- Colautti, R. I., and S. C. H. Barrett. 2013. "Rapid Adaptation to Climate Facilitates Range Expansion of an Invasive Plant." *Science* 342: 364–66.
- Coley, P. D., and J. A. Barone. 1996. "Herbivory and Plant Defenses in Tropical Forests." *Annual Review of Ecology and Systematics* 27: 305–335.
- Coley, P. D., J. P. Bryant, and F. S. Chapin. 1985. "Resource Availability and Plant Antiherbivore Defense." *Science* 230: 895–99.
- Corkill, L. D. 1940. "Cyanogenesis in White Clover (*Trifolium repens* L.). I. Cyanogenesis in Single Plants." *New Zealand Journal of Science and Technology* 22: 65–67.
- Costan, C.-A., W. K. Godsoe, J. L. Bufford, J. W. M. Marris, and P. E. Hulme. 2022. "Can the Enemy Release Hypothesis Explain the Success of *Rumex* (Polygonaceae) Species in an Introduced Range?" *Biological Invasions* 24: 2665–77.
- Daday, H. 1954a. "Gene Frequencies in Wild Populations of *Trifolium repens*. I. Distribution by Latitude." *Heredity* 8: 61–78.
- Daday, H. 1954b. "Gene Frequencies in Wild Populations of *Trifolium repens*. II. Distribution by Altitude." *Heredity* 8: 377–384.
- Daday, H. 1958. "Gene Frequencies in Wild Populations of *Trifolium repens*. III. World Distribution." *Heredity* 12: 169–184.
- Daday, H. 1965. "Gene Frequencies in Wild Populations of *Trifolium repens*. IV. Mechanism of Natural Selection." *Heredity* 20: 355–365.
- de Villemereuil, P., M. Mouterde, O. E. Gaggiotti, and I. Till-Bottraud. 2018. "Patterns of Phenotypic Plasticity and Local Adaptation in the Wide Elevation Range of the Alpine Plant *Arabis alpina*." *Journal of Ecology* 106: 1952–71.
- Dirzo, R., and J. L. Harper. 1982. "Experimental Studies on Slug-Plant Interactions: III. Differences in the Acceptability of Individual Plants of *Trifolium repens* to Slugs and Snails." *Journal of Ecology* 70: 101–117.
- Dlugosch, K. M., and I. M. Parker. 2008a. "Invading Populations of an Ornamental Shrub Show Rapid Life History Evolution Despite Genetic Bottlenecks." *Ecology Letters* 11: 701–9.
- Dlugosch, K. M., and I. M. Parker. 2008b. "Founding Events in Species Invasions: Genetic Variation, Adaptive Evolution, and the Role of Multiple Introductions." *Molecular Ecology* 17: 431–449.
- Dostálek, T., M. B. Rokaya, and Z. Münzbergová. 2020. "Plant Palatability and Trait Responses to Experimental Warming." *Scientific Reports* 10: 10526.
- Easlon, H. M., and A. J. Bloom. 2014. "Easy Leaf Area: Automated Digital Image Analysis for Rapid and Accurate Measurement of Leaf Area." *Applications in Plant Sciences* 2: 1400033.
- Ehrlich, P. R., and P. H. Raven. 1964. "Butterflies and Plants: A Study in Coevolution." *Evolution* 18: 586–608.
- Erfmeier, A. 2013. "Constraints and Release at Different Scales – The Role of Adaptation in Biological Invasions." *Basic and Applied Ecology* 14: 281–88.
- Escudero, A., J. M. Iriando, J. M. Olano, A. Rubio, and R. C. Somolinos. 2000. "Factors Affecting the Establishment of a Gypsophyte: The Case of *Lepidium Subulatum* (Brassicaceae)." *American Journal of Botany* 87: 861–871.
- Estoup, A., V. Ravigné, R. Hufbauer, R. Vitalis, M. Gautier, and B. Facon. 2016. "Is there a Genetic Paradox of Biological Invasion?" *Annual Review of Ecology, Evolution, and Systematics* 47: 51–72.
- Exposito-Alonso, M., 500 Genomes Field Experiment Team, H. A. Burbano, O. Bossdorf, R. Nielsen, and D. Weigel. 2019. "Natural Selection on the *Arabidopsis Thaliana* Genome in Present and Future Climates." *Nature* 573: 126–29.
- Fadoul, H. E., L. J. Albano, M. E. Bergman, M. A. Phillips, and M. T. J. Johnson. 2023. "Assessing the Benefits and Costs of the Hydrogen Cyanide Antiherbivore Defense in *Trifolium repens*." *Plants* 12: 1213.
- Fick, S. E., and R. J. Hijmans. 2017. "WorldClim 2: New 1-km Spatial Resolution Climate Surfaces for Global Land Areas." *International Journal of Climatology* 37: 4302–15.
- Fine, P. V. A., Z. J. Miller, I. Mesones, S. Irazuzta, H. M. Appel, M. H. H. Stevens, I. Sääksjärvi, J. C. Schultz, and P. D. Coley. 2006. "The Growth-Defense Trade-Off and Habitat Specialization by Plants in Amazonian Forests." *Ecology* 87: S150–S162.
- Fournier-Level, A., A. Korte, M. D. Cooper, M. Nordborg, J. Schmitt, and A. M. Wilczek. 2011. "A Map of Local Adaptation in *Arabidopsis thaliana*." *Science* 334: 86–89.
- Fox, J., and S. Weisberg. 2019. *An R Companion to Applied Regression*, 3rd ed. Thousand Oaks, CA: Sage.
- Fugère, V., and A. P. Hendry. 2018. "Human Influences on the Strength of Phenotypic Selection." *Proceedings of the National Academy of Sciences* 115: 10070–75.
- Genton, B. J., P. M. Kotanen, P.-O. Cheptou, C. Adolphe, and J. A. Shykoff. 2005. "Enemy Release but No Evolutionary Loss of Defence in a Plant Invasion: An Inter-Continental Reciprocal Transplant Experiment." *Oecologia* 146: 404–414.
- Gillespie, J. H., and M. Turelli. 1989. "Genotype-Environment Interactions and the Maintenance of Polygenic Variation." *Genetics* 121: 129–138.
- Gleadow, R. M., and I. E. Woodrow. 2002. "Constraints on Effectiveness of Cyanogenic Glycosides in Herbivore Defense." *Journal of Chemical Ecology* 28: 1301–13.
- Griffiths, A. G., R. Moraga, M. Tausen, V. Gupta, T. P. Bilton, M. A. Campbell, R. Ashby, et al. 2019. "Breaking Free: The Genomics of Allopolyploidy-Facilitated Niche Expansion in White Clover." *The Plant Cell* 31: 1466–87.
- Hahn, P. G., A. A. Agrawal, K. I. Sussman, and J. L. Maron. 2019. "Population Variation, Environmental Gradients, and the Evolutionary Ecology of Plant Defense against Herbivory." *The American Naturalist* 193: 20–34.
- Halbritter, A. H., R. Billeter, P. J. Edwards, and J. M. Alexander. 2015. "Local Adaptation at Range Edges: Comparing Elevation

- and Latitudinal Gradients." *Journal of Evolutionary Biology* 28: 1849–60.
- Hargreaves, A. L., R. M. Germain, M. Bontrager, J. Persi, and A. L. Angert. 2020. "Local Adaptation to Biotic Interactions: A Meta-Analysis across Latitudes." *The American Naturalist* 195: 395–411.
- Hendrickson, B. T., C. Stamps, C. M. Patterson, H. Strickland, M. Foster, L. J. Albano, A. Y. Kim, P. Y. Kim, and N. J. Kooyers. 2025. "Evolution of Drought Resistant Strategies Following the Introduction of White Clover (*Trifolium repens* L.)." *Annals of Botany* 135: 1377–92.
- Hendry, A. P. 2017. *Eco-Evolutionary Dynamics*. Princeton, NJ: Princeton University Press.
- Hereford, J. 2009. "A Quantitative Survey of Local Adaptation and Fitness Trade-Offs." *The American Naturalist* 173: 579–588.
- Herms, D. A., and W. J. Mattson. 1992. "The Dilemma of Plants: To Grow or Defend." *The Quarterly Review of Biology* 67: 283–335.
- Hoffmann, A. A., and C. M. Sgrò. 2011. "Climate Change and Evolutionary Adaptation." *Nature* 470: 479–485.
- Hughes, M. A. 1991. "The Cyanogenic Polymorphism in *Trifolium repens* L. (White Clover)." *Heredity* 66: 105–115.
- Hulme, P. E. 2008. "Phenotypic Plasticity and Plant Invasions: Is It All Jack?" *Functional Ecology* 22: 3–7.
- Innes, S., J. S. Santangelo, N. J. Kooyers, K. M. Olsen, and M. T. J. Johnson. 2022. "Evolution in Response to Climate in the Native and Introduced Ranges of a Globally Distributed Plant." *Evolution* 76: 1495–1511.
- Jeschke, J. M., and P. Pyšek. 2018. "Tens Rule." In *Invasion Biology: Hypotheses and Evidence*, edited by J. M. Jeschke and T. Heger. Wallingford, UK: CAB International.
- Johnson, M. T. J., and S. Rasmann. 2011. "The Latitudinal Herbivory-Defence Hypothesis Takes a Detour on the Map." *New Phytologist* 191: 589–592.
- Johnson, M. T. J., J. A. Bertrand, and M. M. Turcotte. 2016. "Precision and Accuracy in Quantifying Herbivory." *Ecological Entomology* 41: 112–121.
- Kakes, P. 1989. "An Analysis of the Costs and Benefits of the Cyanogenic System in *Trifolium repens* L." *Theoretical and Applied Genetics* 77: 111–18.
- Kassambara, A., and F. Mundt. 2020. "Factoextra: Extract and Visualize the Results of Multivariate Data Analyses." R Package Version 1.0.7. <https://CRAN.R-project.org/package=factoextra>
- Kawecki, T. J., and D. Ebert. 2004. "Conceptual Issues in Local Adaptation." *Ecology Letters* 7: 1225–41.
- Keane, R. M., and M. J. Crawley. 2002. "Exotic Plant Invasions and the Enemy Release Hypothesis." *Trends in Ecology & Evolution* 17: 164–170.
- Kjærgaard, T. 2003. "A Plant that Changed the World: The Rise and Fall of Clover 1000–2000." *Landscape Research* 28: 41–49.
- Kolar, C. S., and D. M. Lodge. 2001. "Progress in Invasion Biology: Predicting Invaders." *Trends in Ecology & Evolution* 16: 199–204.
- Kooyers, N. J., A. B. Greenlee, J. M. Colicchio, M. Oh, and B. K. Blackman. 2015. "Replicate Altitudinal Clines Reveal that Evolutionary Flexibility Underlies Adaptation to Drought Stress in Annual *Mimulus guttatus*." *New Phytologist* 206: 152–165.
- Kooyers, N. J., and K. M. Olsen. 2012. "Rapid Evolution of an Adaptive Cyanogenesis Cline in Introduced North American White Clover (*Trifolium repens* L.)." *Molecular Ecology* 21: 2455–68.
- Kooyers, N. J., and K. M. Olsen. 2013. "Searching for the Bull's Eye: Agents and Targets of Selection Vary among Geographically Disparate Cyanogenesis Clines in White Clover (*Trifolium repens* L.)." *Heredity* 111: 495–504.
- Kooyers, N. J., B. Hartman Bakken, M. C. Ungerer, and K. M. Olsen. 2018. "Freeze-Induced Cyanide Toxicity Does Not Maintain the Cyanogenesis Polymorphism in White Clover (*Trifolium repens*)." *American Journal of Botany* 105: 1224–31.
- Kooyers, N. J., J. M. Colicchio, A. B. Greenlee, E. Patterson, N. T. Handloser, and B. K. Blackman. 2019. "Lagging Adaptation to Climate Supersedes Local Adaptation to Herbivory in an Annual Monkeyflower." *The American Naturalist* 194: 541–557.
- Kooyers, N. J., L. R. Gage, A. Al-Lozi, and K. M. Olsen. 2014. "Aridity Shapes Cyanogenesis Cline Evolution in White Clover (*Trifolium repens* L.)." *Molecular Ecology* 23: 1053–70.
- Kuo, W.-H., L. Zhong, S. J. Wright, D. M. Goad, and K. M. Olsen. 2024. "Beyond Cyanogenesis: Temperature Gradients Drive Environmental Adaptation in North American White Clover (*Trifolium repens* L.)." *Molecular Ecology* 33: e17484.
- Lau, J. A., and C. P. terHorst. 2015. "Causes and Consequences of Failed Adaptation to Biological Invasions: The Role of Ecological Constraints." *Molecular Ecology* 24: 1987–98.
- Lee, C.-R., and T. Mitchell-Olds. 2011. "Quantifying Effects of Environmental and Geographical Factors on Patterns of Genetic Differentiation." *Molecular Ecology* 20: 4631–42.
- Leger, E. A., and K. J. Rice. 2007. "Assessing the Speed and Predictability of Local Adaptation in Invasive California Poppies (*Eschscholzia californica*)." *Journal of Evolutionary Biology* 20: 1090–1103.
- Leimu, R., and M. Fischer. 2008. "A Meta-Analysis of Local Adaptation in Plants." *PLoS One* 3: e4010.
- Lemoine, N. P., W. A. Drews, D. E. Burkepile, and J. D. Parker. 2013. "Increased Temperature Alters Feeding Behaviour of a Generalist Herbivore." *Oikos* 122: 1669–78.
- Lenth, R. 2024. "Emmeans: Estimated Marginal Means, Aka Least-Squares Means." R Package Version 1.10.4. <https://rvinth.github.io/emmeans/>
- Li, X.-M., D.-Y. She, D.-Y. Zhang, and W.-J. Liao. 2015. "Life History Trait Differentiation and Local Adaptation in Invasive Populations of *Ambrosia artemisiifolia* in China." *Oecologia* 177: 669–677.
- Lucas, M. S., I. Hensen, C. D. Barratt, R. M. Callaway, W. Durka, Y. Lekberg, D. U. Nagy, et al. 2024. "Re-Focusing Sampling, Design and Experimental Methods to Assess Rapid Evolution by Non-Native Plant Species." *Biological Invasions* 26: 1327–43.
- Macel, M., C. S. Lawson, S. R. Mortimer, M. Šmilauerová, A. Bischoff, I. Crémieux, J. Doležal, et al. 2007. "Climate vs. Soil Factors in Local Adaptation of Two Common Plant Species." *Ecology* 88: 424–433.
- Machingura, M., E. Salomon, J. M. Jez, and S. D. Ebbs. 2016. "The β -Cyanoalanine Synthase Pathway: Beyond Cyanide Detoxification." *Plant, Cell and Environment* 39: 2329–41.

- Maron, J. L., M. Vilà, R. Bommarco, S. Elmendorf, and P. Beardsley. 2004. "Rapid Evolution of an Invasive Plant." *Ecological Monographs* 74: 261–280.
- Massad, T. J., and L. A. Dyer. 2010. "A Meta-Analysis of the Effects of Global Environmental Change on Plant-Herbivore Interactions." *Arthropod-Plant Interactions* 4: 181–88.
- Miles, L. S., S. T. Breitbart, H. H. Wagner, and M. T. J. Johnson. 2019. "Urbanization Shapes the Ecology and Evolution Plant-Arthropod Herbivore Interactions." *Frontiers in Ecology and Evolution* 7: 310.
- Mitchell, C. E., A. A. Agrawal, J. D. Bever, G. S. Gilbert, R. A. Huffbauer, J. N. Klironomos, J. L. Maron, et al. 2006. "Biotic Interactions and Plant Invasions." *Ecology Letters* 9: 726–740.
- Moles, A. T., and M. Westoby. 2004. "Seedling Survival and Seed Size: A Synthesis of the Literature." *Journal of Ecology* 92: 372–383.
- Møller, B. L. 2010. "Functional Diversifications of Cyanogenic Glucosides." *Current Opinion in Plant Biology* 13: 337–346.
- Moreira, X., B. Castagneyrol, L. Abdala-Roberts, J. C. Berny-Mier y Teran, B. G. H. Timmermans, H. H. Bruun, F. Covelo, G. Glauser, S. Rasman, and A. J. M. Tack. 2018. "Latitudinal Variation in Plant Chemical Defences Drives Latitudinal Patterns of Leaf Herbivory." *Ecography* 41: 1124–34.
- Müller-Schärer, H., U. Schaffner, and T. Steinger. 2004. "Evolution in Invasive Plants: Implications for Biological Control." *Trends in Ecology & Evolution* 19: 417–422.
- Murray-Stoker, D., and M. T. J. Johnson. 2024. "Mosaic of Local Adaptation between White Clover and Rhizobia along an Urbanization Gradient." *Journal of Ecology* 112: 1150–63.
- Nagy, D. U., A. E. Thoma, M. Al-Gharaibeh, R. M. Callaway, S. L. Flory, L. J. Frazee, M. Hartmann, et al. 2024. "Among-Population Variation in Drought Responses Is Consistent across Life Stages but Not between Native and Non-Native Ranges." *New Phytologist* 243: 922–935.
- O'Donnell, M. S., and D. A. Ignizio. 2012. "Bioclimatic Predictors for Supporting Ecological Applications in the Conterminous United States." *US Geological Survey Data Series* 10: 4–9.
- Oduor, A. M. O., R. Leimu, and M. van Kleunen. 2016. "Invasive Plant Species Are Locally Adapted Just as Frequently and at Least as Strongly as Native Plant Species." *Journal of Ecology* 104: 957–968.
- Olsen, K. M., and L. L. Small. 2018. "Micro- and Macroevolutionary Adaptation though Repeated Loss of a Complete Metabolic Pathway." *New Phytologist* 219: 757–766.
- Olsen, K. M., B. L. Sutherland, and L. L. Small. 2007. "Molecular Evolution of the *Li/li* Chemical Defence Polymorphism in White Clover (*Trifolium repens* L.)." *Molecular Ecology* 16: 4180–93.
- Olsen, K. M., D. M. Goad, S. J. Wright, M. L. Dutta, S. R. Myers, L. L. Small, and L.-F. Li. 2021. "Dual-Species Origin of an Adaptive Chemical Defense Polymorphism." *New Phytologist* 232: 1477–87.
- Pal, R. W., J. L. Maron, D. U. Nagy, L. P. Waller, A. Tosto, H. Liao, and R. M. Callaway. 2020. "What Happens in Europe Stays in Europe: Apparent Evolution by an Invader Does Not Help at Home." *Ecology* 101: e3072.
- Pannell, J. R., J. R. Auld, Y. Brandvain, M. Burd, J. W. Busch, P.-O. Cheptou, J. K. Connor, et al. 2015. "The Scope of Baker's Law." *New Phytologist* 208: 656–667.
- Parmesan, C. 2006. "Ecological and Evolutionary Responses to Recent Climate Change." *Annual Review of Ecology, Evolution, and Systematics* 37: 637–669.
- Pickett, S. T. A. 1989. "Space-for-Time Substitution as an Alternative for Long-Term Studies." In *Long-Term Studies in Ecology: Approaches and Alternatives*, edited by G. E. Likens. New York, NY: Springer.
- Posit Team. 2025. *RStudio: Integrated Development Environment for R*. Boston, MA: Posit Software, PBC.
- Pyšek, P., P. E. Hulme, D. Simberloff, S. Bacher, T. M. Blackburn, J. T. Carlton, W. Dawson, et al. 2020. "Scientists' Warning on Invasive Alien Species." *Biological Reviews* 95: 1511–34.
- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing.
- Saikkonen, K., K. Taulavuori, T. Hyvönen, P. E. Gundel, C. E. Hamilton, I. Vänninen, A. Nissinen, and M. Helander. 2012. "Climate Change-Driven Species' Range Shifts Filtered by Photoperiodism." *Nature Climate Change* 2: 239–242.
- Samis, K. E., J. R. Stinchcombe, and C. J. Murren. 2019. "Population Climatic History Predicts Phenotypic Responses in Novel Environments for *Arabidopsis thaliana* in North America." *American Journal of Botany* 106: 1068–80.
- Santangelo, J. S., R. W. Ness, B. Cohan, C. R. Fitzpatrick, S. G. Innes, S. Koch, L. S. Miles, et al. 2022. "Global Urban Environmental Change Drives Adaptation in White Clover." *Science* 375: 1275–81.
- Savolainen, O., M. Lascoux, and J. Merilä. 2013. "Ecological Genomics of Local Adaptation." *Nature Reviews Genetics* 14: 807–820.
- Sheng, M., C. Rosche, M. Al-Gharaibeh, L. S. Bullington, R. M. Callaway, T. Clark, C. C. Cleveland, et al. 2022. "Acquisition and Evolution of Enhanced Mutualism—An Underappreciated Mechanism for Invasive Success?" *The ISME Journal* 16: 2467–78.
- Siepielski, A. M., M. B. Morrissey, M. Buoro, S. M. Carlson, C. M. Caruso, S. Clegg, T. Coulson, et al. 2017. "Precipitation Drives Global Variation in Natural Selection." *Science* 355: 959–962.
- Simón-Porcar, V. I., J. L. Silva, and M. Vallejo-Marín. 2021. "Rapid Local Adaptation in both Sexual and Asexual Invasive Populations of Monkeyflowers (*Mimulus* spp.)." *Annals of Botany* 127: 655–668.
- St Clair, J. B., and G. T. Howe. 2007. "Genetic Maladaptation of Coastal Douglas-Fir Seedlings to Future Climates." *Global Change Biology* 13: 1441–54.
- Taylor, C. M., and A. Hastings. 2005. "Allee Effects in Biological Invasions." *Ecology Letters* 8: 895–908.
- Thompson, K. A., M. Renaudin, and M. T. J. Johnson. 2016. "Urbanization Drives the Evolution of Parallel Clines in Plant Populations." *Proceedings of the Royal Society B: Biological Sciences* 283: 20162180.
- Till, I. 1987. "Variability of Expression of Cyanogenesis in White Clover (*Trifolium repens* L.)." *Heredity* 59: 265–271.
- Toräng, P., J. Wunder, J. T. Obeso, M. Herzog, G. Coupland, and J. Ågren. 2015. "Large-Scale Adaptive Differentiation in the Alpine Perennial *Arabis alpina*." *New Phytologist* 206: 459–470.
- Tsutsui, N. D., A. V. Suarez, D. A. Holway, and T. J. Case. 2000. "Reduced Genetic Variation and the Success of an Invasive Species." *Proceedings of the National Academy of Sciences* 97: 5948–53.

- Turesson, G. 1922. "The Genotypical Response of the Plant Species to the Habitat." *Hereditas* 3: 211–350.
- Valladares, F., S. Matesanz, F. Guilhaumon, M. B. Araújo, L. Balaguer, M. Benito-Garzon, W. Cornwell, et al. 2014. "The Effects of Phenotypic Plasticity and Local Adaptation on Forecasts of Species Range Shifts under Climate Change." *Ecology Letters* 17: 1351–64.
- van Boheemen, L. A., D. Z. Atwater, and K. A. Hodgins. 2019. "Rapid and Repeated Local Adaptation to Climate in an Invasive Plant." *New Phytologist* 222: 614–627.
- van Boheemen, L. A., E. Lombaert, K. A. Nurkowski, B. Gauffre, L. H. Rieseberg, and K. A. Hodgins. 2017. "Multiple Introductions, Admixture and Bridgehead Invasion Characterize the Introduction History of *Ambrosia artemisiifolia* in Europe and Australia." *Molecular Ecology* 26: 5421–34.
- Walther, G.-R., A. Roques, P. E. Hulme, M. T. Sykes, P. Pyšek, I. Kühn, M. Zobel, et al. 2009. "Alien Species in a Warmer World: Risks and Opportunities." *Trends in Ecology & Evolution* 24: 686–693.
- Wang, T., A. Hamann, D. Spittlehouse, and C. Carroll. 2016. "Locally Downscaled and Spatially Customizable Climate Data for Historical and Future Periods for North America." *PLoS One* 11: e0156720.
- Wilczek, A. M., M. D. Cooper, T. M. Korves, and J. Schmitt. 2014. "Lagging Adaptation to Warming Climate in *Arabidopsis thaliana*." *Proceedings of the National Academy of Sciences* 111: 7906–13.
- Wolkovich, E. M., T. J. Davies, H. Schaefer, E. E. Cleland, B. I. Cook, S. E. Travers, C. G. Willis, and C. C. Davis. 2013. "Temperature-Dependent Shifts in Phenology Contribute to the Success of Exotic Species with Climate Change." *American Journal of Botany* 100: 1407–21.
- Wright, S. J., D. Cui Zhou, A. Kuhle, and K. M. Olsen. 2018. "Continent-Wide Climatic Variation Drives Local Adaptation in North American White Clover." *Journal of Heredity* 109: 78–89.
- Wright, S. J., D. M. Goad, B. L. Gross, P. R. Muñoz, and K. M. Olsen. 2022. "Genetic Trade-Offs Underlie Divergent Life Strategies for Local Adaptation in White Clover." *Molecular Ecology* 31: 3742–60.
- Züst, T., C. Heichinger, U. Grossniklaus, R. Harrington, D. J. Kliebenstein, and L. A. Turnbull. 2012. "Natural Enemies Drive Geographic Variation in Plant Defenses." *Science* 338: 116–19.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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